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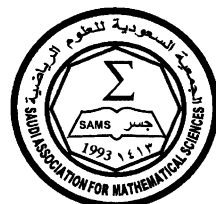
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ON SUBMANIFOLDS OF CODIMENSION TWO OF A NEARLY TRANS-SASAKIAN MANIFOLD

U.C. DE, A. K. SENGUPTA, RAJENDRA PRASAD AND MUKUT MANI TRIPATHI

ABSTRACT. Some sufficient conditions for the submanifold of codimension two of a nearly trans-Sasakian manifold, with trivial normal bundle, to admit a nearly trans-Sasakian structure are obtained.

1. INTRODUCTION

In [12], S. Tanno classified connected almost contact metric manifolds whose automorphism groups possess the maximum dimension. For such a manifold, the sectional curvature of plane sections containing ξ is a constant, say c . He showed that they can be divided into three classes: **(1)** homogeneous normal contact Riemannian manifolds with $c > 0$, **(2)** global Riemannian products of a line or a circle with a Kaehler manifold of constant holomorphic sectional curvature if $c = 0$ and **(3)** a warped product space $\mathbb{R} \times_f \mathbb{C}^n$ if $c < 0$. It is known that the manifolds of class **(1)** are characterized by admitting a Sasakian structure. The manifolds of class **(2)** are characterized by a tensorial relation admitting a cosymplectic structure. Kenmotsu ([8]) characterized the differential geometric properties of the manifolds of class **(3)**; the structure so obtained is now known as Kenmotsu structure. In general, these structures are not Sasakian ([8]).

In the Gray-Hervella classification of almost Hermitian manifolds ([5]), there appears a class, $\mathcal{W}_4$, of Hermitian manifolds which are closely related to locally conformal Kaehler manifolds. An almost contact metric structure on a manifold $\bar{M}$ is called a *trans-Sasakian structure* ([11]) if the product manifold

$\bar{M} \times \mathbb{R}$ belongs to the class $\mathcal{W}_4$. The class $\mathcal{C}_6 \oplus \mathcal{C}_5$ ([10]) coincides with the class of trans-Sasakian structures of type (α, β) . We note that trans-Sasakian structures of type $(0, 0)$ are cosymplectic ([1]), trans-Sasakian structures of type $(0, \beta)$ are β -Kenmotsu ([6]) and trans-Sasakian structures of type $(\alpha, 0)$ are α -Sasakian ([6]).

Recently, C. Gherghe ([4]) introduced a nearly trans-Sasakian structure of type (α, β) , which generalizes trans-Sasakian structure in the same sense as nearly Sasakian structures generalize Sasakian ones. A trans-Sasakian structure is always a nearly trans-Sasakian structure. Moreover, a nearly trans-Sasakian structure of type (α, β) is nearly-Sasakian ([3]) or nearly Kenmotsu ([13]) or nearly cosymplectic ([1]) according as $\beta = 0$ or $\alpha = 0$ or $\alpha = 0 = \beta$.

In the present paper, we study the submanifolds of codimension two of nearly trans-Sasakian manifolds. Section 2 contains necessary details about nearly trans-Sasakian structures and a submanifold of codimension two of an almost contact metric manifold. Definition of nearly trans-Sasakian manifolds is given. It is shown that a submanifold of codimension two of an almost contact metric manifold admits a $(f, g, U_{(k)}, u_{(k)}, a_{(k)})$ structure ([9], [14]) and under certain conditions this structure yields an almost contact metric structure on the submanifold ([7]). In the last section 3, in the main theorem of the paper, we establish that under certain conditions the induced almost contact metric structure on a submanifold of codimension two of a nearly trans-Sasakian manifold is also nearly trans-Sasakian. Finally, similar results are stated for some particular types of almost contact metric structures which are nearly trans-Sasakian: nearly Sasakian, nearly Kenmotsu, nearly cosymplectic, trans-Sasakian, Kenmotsu and cosymplectic.

2. NEARLY TRANS-SASAKIAN MANIFOLDS

Let $\bar{M}$ be an almost contact metric manifold ([1]) with an almost contact metric structure (ϕ, ξ, η, g) , that is, ϕ is a $(1, 1)$ tensor field, ξ is a vector field; η is 1-form and g is a compatible Riemannian metric such that

$$(1) \quad \phi^2 = -I + \eta \otimes \xi, \quad \eta(\xi) = 1, \quad \phi(\xi) = 0, \quad \eta \circ \phi = 0,$$

$$(2) \quad g(\phi X, \phi Y) = g(X, Y) - \eta(X)\eta(Y),$$

$$(3) \quad g(X, \phi Y) = -g(\phi X, Y), \quad g(X, \xi) = \eta(X)$$

for all $X, Y \in T\bar{M}$.

An almost contact metric structure (ϕ, ξ, η, g) on $\bar{M}$ is called a *trans-Sasakian structure* (Oubina, [11]) if $(\bar{M} \times \mathbb{R}, J, G)$ belongs to the class $\mathcal{W}_4$ ([5]), where J is the almost complex structure on $\bar{M} \times \mathbb{R}$ defined by

$$J(X, ad/dt) = (\phi X - a\xi, \eta(X)d/dt)$$

for all vector fields X on $\bar{M}$ and smooth functions a on $\bar{M} \times \mathbb{R}$ and G is the product metric on $\bar{M} \times \mathbb{R}$. This may be expressed by the condition (Blair and Oubina [2])

$$(4) \quad (\bar{\nabla}_X \phi)Y = \alpha(g(X, Y)\xi - \eta(Y)X) + \beta(g(\phi X, Y)\xi - \eta(Y)\phi X)$$

for some smooth functions α and β on $\bar{M}$, and we say that the trans-Sasakian structure is of type (α, β) .

An almost contact metric structure (ϕ, ξ, η, g) on $\bar{M}$ is called a *nearly trans-Sasakian structure* ([4]) if

$$(5) \quad \begin{aligned} (\bar{\nabla}_X \phi)Y + (\bar{\nabla}_Y \phi)X &= \alpha(2g(X, Y)\xi - \eta(Y)X - \eta(X)Y) \\ &\quad - \beta(\eta(Y)\phi X + \eta(X)\phi Y). \end{aligned}$$

A trans-Sasakian structure is always nearly trans-Sasakian.

Let M be a submanifold of codimension two of an almost contact metric manifold $\bar{M}$ endowed with an almost contact metric structure (ϕ, ξ, η, g) . We identify M with $i(M)$ in $\bar{M}$, where $i : M \rightarrow \bar{M}$ is the imbedding map. We assume that there is a normal frame $\{N_1, N_2\}$ defined globally over the submanifold M (that is, the normal plane bundle is trivial). Then we have the decomposition law for ξ , $\phi i_* X$ and ϕN_r ($r = 1, 2$) with respect to the frame $\{N_1, N_2\}$ as follows:

$$(6) \quad \xi = i_* U_3 - a_2 N_1 - a_1 N_2,$$

$$(7) \quad \phi i_* X = i_* fX + u_1(X) N_1 + u_2(X) N_2,$$

$$(8) \quad \phi N_1 = -i_* U_1 + a_3 N_2,$$

$$(9) \quad \phi N_2 = -i_* U_2 - a_3 N_1,$$

where f denotes a $(1, 1)$ tensor field, U_r , $r = 1, 2, 3$ three vector fields, u_r , $r = 1, 2$, two 1-forms, a_r , $r = 1, 2, 3$, three scalar fields on M . The induced Riemannian metric g' on M is given by

$$(10) \quad g'(X, Y) = g(X, Y), \quad X, Y \in TM.$$

We define a 1-form u_3 on M by

$$(11) \quad u_3(X) = \eta(i_*X), \quad X, Y \in TM.$$

Then one obtains the following equations :

$$(12) \quad f^2 = -I + u_1 \otimes U_1 + u_2 \otimes U_2 + u_3 \otimes U_3,$$

$$(13) \quad \left. \begin{aligned} fU_1 &= -a_3U_2 + a_2U_3, & u_1 \circ f &= a_3u_2 - a_2u_3, \\ fU_2 &= a_1U_3 + a_3U_1, & u_2 \circ f &= -a_1u_3 - a_3u_1, \\ fU_3 &= -a_2U_1 - a_1U_2, & u_3 \circ f &= a_2u_1 + a_1u_2, \end{aligned} \right\}$$

$$(14) \quad \left. \begin{aligned} g'(X, fY) &= -g'(fX, Y), \\ g'(X, U_i) &= u_i(X), \\ g'(fX, fY) &= g'(X, Y) - \sum_i u_i(X) u_i(Y) \end{aligned} \right\}$$

$$(15) \quad \left. \begin{aligned} g'(U_1, U_1) &= |U_1|^2 = 1 - a_2^2 - a_3^2, \\ g'(U_1, U_2) &= u_1(U_2) = u_2(U_1) = -a_1a_2, \\ g'(U_2, U_2) &= |U_2|^2 = 1 - a_3^2 - a_1^2, \\ g'(U_2, U_3) &= u_2(U_3) = u_3(U_2) = a_2a_3, \\ g'(U_3, U_3) &= |U_3|^2 = 1 - a_1^2 - a_2^2, \\ g'(U_3, U_1) &= u_3(U_1) = u_1(U_3) = -a_3a_1. \end{aligned} \right\}$$

Thus a submanifold of codimension two of an almost contact metric manifold admits a $(f, g, U_{(k)}, u_{(k)}, a_{(k)})$ structure given in [9] (See also [14]). If $\varkappa$ is a scalar field defined by $\varkappa = \sqrt{a_2^2 + a_1^2 + a_3^2}$ on the submanifold of codimension two of an almost contact metric manifold, then $\varkappa$ is independent of a choice of the normal frame $\{N_1, N_2\}$ and satisfies $0 \leq \varkappa \leq 1$ ([7]).

Defining

$$(16) \quad U = a_3U_3 - a_1U_1 + a_2U_2, \quad u = a_3u_3 - a_1u_1 + a_2u_2,$$

in view of (12)-(15) it follows that

$$(17) \quad \left. \begin{aligned} u(U) &= \varkappa^2, \quad u(X) = g'(X, U), \quad fU = 0, \quad u \circ f = 0, \\ f^3X + fX &= u_1(fX)fU_1 + u_2(fX)fU_2 + u_3(fX)fU_3, \\ f^4X + (\varkappa^2 + 1)f^2X + \varkappa^2X &= u(X)U, \\ f^5X + (\varkappa^2 + 1)f^3X + \varkappa^2fX &= 0 \end{aligned} \right\}$$

for all $X \in TM$.

For a submanifold of codimension two of an almost contact metric manifold to admit an almost contact metric structure, it is sufficient that a condition ([7])

$$\text{either } \varkappa = 0 \text{ on } M \quad \text{or} \quad 0 < \varkappa \leq 1 \text{ on } M.$$

Thus if the scalar field $\varkappa$ ($0 \leq \varkappa \leq 1$) is constant on M , then an almost contact metric structure can always be induced on M . In case of $0 < \varkappa \leq 1$, defining

$$(18) \quad f' = \frac{1}{\varkappa(\varkappa+1)} (f^3 + (\varkappa^2 + \varkappa + 1) f), \quad u' = \frac{1}{\varkappa} u, \quad U' = \frac{1}{\varkappa} U$$

one can verify that (f', u', U', g') is an almost contact metric structure on M .

3. SUBMANIFOLDS OF CODIMENSION TWO OF NEARLY TRANS-SASAKIAN MANIFOLDS

For the submanifold M of codimension two of an almost contact metric manifold, we have Gauss and Weingarten formulae as

$$(19) \quad \bar{\nabla}_{i_*X} i_*Y = i_*\nabla_X Y + h_1(X, Y) N_1 + h_2(X, Y) N_2,$$

$$(20) \quad \bar{\nabla}_{i_*X} N_1 = -i_*H_1X + w(X) N_2, \quad \bar{\nabla}_{i_*X} N_2 = -i_*H_2X - w(X) N_1,$$

where h_1 and h_2 (resp. H_1 and H_2) are the second fundamental tensor fields of type $(0, 2)$ (resp. $(1, 1)$) and w is the third fundamental 1-form with respect to $\{N_1, N_2\}$. h_1 and h_2 are symmetric and satisfy

$$h_1(X, Y) = g'(H_1X, Y), \quad h_2(X, Y) = g'(H_2X, Y).$$

Now, we assume that $\bar{M}$ is nearly trans-Sasakian; and then for all $X, Y \in TM$, we have

$$(21) \quad \begin{aligned} (\bar{\nabla}_{i_*X} \phi) i_*Y + (\bar{\nabla}_{i_*Y} \phi) i_*X &= \alpha(2g(i_*X, i_*Y)\xi - \eta(i_*Y) i_*X - \eta(i_*X) i_*Y) \\ &\quad - \beta(\eta(i_*Y) \phi i_*X + \eta(i_*X) \phi i_*Y). \end{aligned}$$

By using (7)-(9), (19) and (20), the left hand side of (21) reduces to

$$i_*((\nabla_X f)Y - u_1(Y)H_1X - u_2(Y)H_2X + h_1(X, Y)U_1 + h_2(X, Y)U_2)$$

$$\begin{aligned}
& +i_*((\nabla_Y f)X - u_1(X)H_1Y - u_2(X)H_2Y + h_1(X, Y)U_1 + h_2(X, Y)U_2) \\
& + (h_1(X, fY) + (\nabla_X u_1)Y - w(X)u_2(Y) + a_3h_2(X, Y))N_1 \\
& + (h_1(Y, fX) + (\nabla_Y u_1)X - w(Y)u_2(X) + a_3h_2(X, Y))N_1 \\
& + (h_2(X, fY) + (\nabla_X u_2)Y + w(X)u_1(Y) - a_3h_1(X, Y))N_2 \\
& + (h_2(Y, fX) + (\nabla_Y u_2)X + w(Y)u_1(X) - a_3h_1(X, Y))N_2;
\end{aligned}$$

and using (6), (7), (10) and (11) the right hand side of (21) reduces to

$$\begin{aligned}
& i_*(\alpha(2g'(X, Y)U_3 - u_3(Y)X - u_3(X)Y) - \beta(u_3(Y)fX + u_3(X)fY)) \\
& - (2\alpha a_2g'(X, Y) + \beta u_3(Y)u_1(X) + \beta u_3(X)u_1(Y))N_1 \\
& - (2\alpha a_1g'(X, Y) + \beta u_3(Y)u_2(X) + \beta u_3(X)u_2(Y))N_2.
\end{aligned}$$

Comparing the tangential parts to M in the above two expressions, we obtain

$$\begin{aligned}
& (\nabla_X f)Y + (\nabla_Y f)X \\
& = \alpha(2g'(X, Y)U_3 - u_3(Y)X - u_3(X)Y) - \beta(u_3(Y)fX + u_3(X)fY) \\
& \quad + u_1(Y)H_1X + u_2(Y)H_2X + u_1(X)H_1Y + u_2(X)H_2Y \\
(22) \quad & - 2(g'(H_1X, Y)U_1 + g'(H_2X, Y)U_2)
\end{aligned}$$

for all $X, Y \in TM$.

Now, we present the main theorem.

Theorem 3.1 *Suppose that $\kappa = 1$ on a submanifold M of codimension two of a nearly trans-Sasakian manifold $\bar{M}$ with trivial normal plane bundle. Then the induced almost contact metric structure on M will be nearly trans-Sasakian if the second fundamental tensors H_1 and H_2 are related by $a_1H_1 = a_2H_2$.*

Proof. Since $1 = \kappa^2 = a_2^2 + a_1^2 + a_3^2$, therefore in view of Lemma 1.1 of [9], U_1, U_2 and U_3 are linearly dependent. Therefore, we have

$$|U_1| \cdot |U_2| = g'(U_1, U_2), \quad |U_2| \cdot |U_3| = g'(U_2, U_3), \quad |U_3| \cdot |U_1| = g'(U_3, U_1).$$

Since $\kappa = 1$ from (18) we have $U = U'$ and $u = u'$. Then it can be easily verified that

$$U_1 = -a_1U, \quad U_2 = a_2U \quad \text{and} \quad U_3 = a_3U,$$

$$u_1 = -a_1u, \quad u_2 = a_2u, \quad u_3 = a_3u.$$

Also, from (17) and (18) it follows that $f' = f$.

Hence, (22) reduces to

$$\begin{aligned} (\nabla_X f)Y + (\nabla_Y f)X &= a_3\alpha(2g'(X, Y)U - u(Y)X - u(X)Y) \\ &\quad - a_3\beta(u(Y)fX + u(X)fY) \\ &\quad - u(Y)(a_1H_1 - a_2H_2)X - u(X)(a_1H_1 - a_2H_2)Y \\ &\quad - 2g'((a_1H_1 - a_2H_2)X, Y)U \end{aligned}$$

for all $X, Y \in TM$. From above equation it follows that the almost contact metric structure (f, u, U, g') on M will be nearly trans-Sasakian if $a_1H_1 = a_2H_2$.

In view of the above theorem, we are able to state the following theorem.

Theorem 3.2 *Suppose that $\kappa = 1$ on a submanifold M of codimension two of an almost contact metric manifold $\bar{M}$ with trivial normal plane bundle. Then the induced almost contact metric structure on M will be nearly Sasakian, nearly Kenmotsu, nearly cosymplectic, trans-Sasakian, Sasakian, Kenmotsu and cosymplectic according as the almost contact metric structure on $\bar{M}$ is nearly Sasakian, nearly Kenmotsu, nearly cosymplectic, trans-Sasakian, Sasakian, Kenmotsu and cosymplectic respectively, provided the second fundamental tensors H_1 and H_2 satisfy $a_1H_1 = a_2H_2$.*

A submanifold M is called ϕ -invariant if $\phi i_*X \in TM$ for all $X \in TM$ or equivalently, if $\phi i_*X = i_*fX$ holds identically. Hence, $a_1 = 0 = a_2$ and $a_3^2 = 1$. Accordingly, we have $\kappa = 1$. Thus we are able to state the following

Theorem 3.3 *Suppose that M is a ϕ -invariant submanifold of codimension two of an almost contact metric manifold $\bar{M}$ with trivial normal plane bundle. Then the induced almost contact metric structure on M will be nearly trans-Sasakian, nearly Sasakian, nearly Kenmotsu, nearly cosymplectic, trans-Sasakian, Sasakian, Kenmotsu and cosymplectic according as the almost contact metric structure on $\bar{M}$ is nearly trans-Sasakian, nearly Sasakian, nearly Kenmotsu, nearly cosymplectic, trans-Sasakian, Sasakian, Kenmotsu and cosymplectic respectively.*

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REMARKS ON A FOUR-DIMENSIONAL COMPACT ALMOST KÄHLER EINSTEIN MANIFOLD

TAKASHI OGURO AND KOUEI SEKIGAWA

ABSTRACT. In this paper, we will prove that the skew-symmetric part of the Ricci *-tensor of a 4-dimensional compact almost Kähler Einstein manifold M vanishes somewhere on M . Further, we will give another proof of the theorem in [1] by V. Apostolov and J. Armstrong.

1. INTRODUCTION

An almost Hermitian manifold $M = (M, J, g)$ is called an almost Kähler manifold if the Kähler form Ω of M defined by $\Omega(X, Y) = g(X, JY)$ is closed. The condition $d\Omega = 0$ is equivalent to $\mathfrak{S}_{X,Y,Z} g((\nabla_X J)Y, Z) = 0$, where $\mathfrak{S}_{X,Y,Z}$ denotes the cyclic sum with respect to X, Y, Z . An almost Kähler manifold with integrable almost complex structure is a Kähler manifold. A non-Kähler, almost Kähler manifold is called a strictly almost Kähler manifold.

Concerning the integrability of almost Kähler manifolds, the following conjecture by S.I. Goldberg is well known:

Conjecture. *A compact almost Kähler Einstein manifold is integrable.*

This conjecture is true in the case where the scalar curvature of M is non-negative ([8]), and many progresses have been made by many authors under certain additional curvature conditions. However, the conjecture is still open. Recently, the authors have proved the following.

Theorem A ([6]) *Let $M = (M, J, g)$ be a four-dimensional compact almost Kähler Einstein manifold. If the norm of skew-symmetric part of the Ricci $\ast$ -tensor is constant, then M is a Kähler manifold.*

In this paper, we shall prove the following theorem which was suggested by T. Draghici in a private communication.

Theorem 1 *Let $M = (M, J, g)$ be a four-dimensional compact almost Kähler Einstein manifold. Then, the skew-symmetric part of the Ricci $\ast$ -tensor vanishes somewhere on M .*

Taking account of the result in [7], we can obtain Theorem A as a corollary of Theorem 1. Further, in section , we will give another proof of the following theorem by V. Apostolov and J. Armstrong.

Theorem 2 ([1]) *A four-dimensional compact almost Kähler Einstein manifold M with Hermitian Weyl tensor is a Kähler manifold.*

In this theorem, it seems interesting to study the integrability without compactness.

2. PRELIMINARIES

In this section, we prepare several fundamental formulas which will be used in our argument. Most of them are established in [6].

Let $M = (M, J, g)$ be a four-dimensional almost Hermitian manifold. The Kähler form of M is defined by $\Omega(X, Y) = g(X, JY)$ for $X, Y \in \mathfrak{X}(M)$, where $\mathfrak{X}(M)$ is the Lie algebra of all smooth vector fields on M . We assume that M is oriented by the volume form $dM = \Omega^2/2$. We denote by ∇ , R , ρ and τ the Riemannian connection, the curvature tensor, the Ricci tensor and the scalar curvature of M respectively, where $R(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X, Y]}$. The Ricci $\ast$ -tensor ρ^* of M is defined by

$$\rho^*(x, y) = \frac{1}{2} \text{trace of } (z \mapsto R(x, Jy)Jz)$$

for $x, y, z \in T_p M$, the tangent space of M at $p \in M$. We see immediately that $\rho^*(x, y) = \rho^*(Jy, Jx)$. Further, it is known that

$$(2.1) \quad \frac{1}{2} \{ \rho(x, y) + \rho(Jx, Jy) \} - \frac{1}{2} \{ \rho^*(x, y) + \rho^*(y, x) \} = -\frac{\tau^* - \tau}{4} g(x, y)$$

holds for $x, y \in T_p M$, $p \in M$ ([4]). The $*$ -scalar curvature τ^* is the trace of the linear endomorphism Q^* of $T_p M$ defined by $g(Q^* x, y) = \rho^*(x, y)$ for $x, y \in T_p M$.

In this paper, for any orthonormal bases (resp. any local orthonormal frame field) $\{e_i\}_{i=1,\dots,4}$ of $T_p M$ (resp. on a neighborhood of p), we shall adopt the following notational convention:

$$\begin{aligned}\Gamma_{ijk} &= g(\nabla_{e_i} e_j, e_k), \quad J_{ij} = g(Je_i, e_j), \quad \nabla_i J_{jk} = g((\nabla_{e_i} J)e_j, e_k), \\ R_{ijkl} &= g(R(e_i, e_j)e_k, e_l), \dots, R_{i\bar{j}\bar{k}\bar{l}} = g(R(Je_i, Je_j)Je_k, Je_l), \\ \rho_{ij} &= \rho(e_i, e_j), \dots, \rho_{i\bar{j}} = \rho(Je_i, Je_j), \\ \rho_{ij}^* &= \rho^*(e_i, e_j), \dots, \rho_{i\bar{j}}^* = \rho^*(Je_i, Je_j),\end{aligned}$$

and so on, where the Latin indices run over the range $1, \dots, 4$. Then, we have $\Gamma_{ijk} = -\Gamma_{ikj}$, $J_{ij} = -J_{ji}$, $\nabla_i J_{jk} = -\nabla_i J_{kj}$, $\nabla_i J_{\bar{j}\bar{k}} = -\nabla_i J_{\bar{k}\bar{j}}$.

We denote by $\wedge^2 M$ the vector bundle of real 2-forms on M . The bundle $\wedge^2 M$ inherits a natural inner product from the Riemannian metric g (also denoted by g). We have the following decomposition of $\wedge^2 M$:

$$(2.2) \quad \wedge^2 M = \mathbb{R}\Omega \oplus \wedge_0^{1,1} M \oplus LM,$$

where $\wedge_0^{1,1} M$ denotes the vector bundle of real J -invariant 2-forms, LM the vector bundle of real J -skew-invariant 2-forms of M which are perpendicular to Ω , respectively. The subbundle LM is endowed with the natural complex structure J defined by $J\Phi(X, Y) = -\Phi(JX, Y)$ for any local section Φ of LM and $X, Y \in \mathfrak{X}(M)$. The subbundle $\mathbb{R}\Omega \oplus LM$ is identified itself with the bundle $\wedge_+^2 M$ of self-dual 2-forms of M , while LM is the bundle $\wedge_-^2 M$ of anti-self-dual 2-forms of M :

$$(2.3) \quad \wedge_+^2 M = \mathbb{R}\Omega \oplus LM,$$

$$(2.4) \quad \wedge_-^2 M = \wedge_0^{1,1} M.$$

Let $\{e_i\} = \{e_1, e_2 = Je_1, e_3, e_4 = Je_3\}$ be any local unitary frame and we put $\{e^i\} = \{\iota(e_i)\}$, where $\iota : TM \rightarrow T^*M$ denotes the duality defined by means of the metric g . Then Ω is given by $\Omega = -e^1 \wedge e^2 - e^3 \wedge e^4$. Further, if we put $\Phi = (e^1 \wedge e^3 - e^2 \wedge e^4)/\sqrt{2}$, $J\Phi = (e^1 \wedge e^4 + e^2 \wedge e^3)/\sqrt{2}$, $\Psi_1 = (e^1 \wedge e^2 - e^3 \wedge e^4)/\sqrt{2}$, $\Psi_2 = (e^1 \wedge e^3 + e^2 \wedge e^4)/\sqrt{2}$, $\Psi_3 = (e^1 \wedge e^4 - e^2 \wedge e^3)/\sqrt{2}$, then $\{\Phi, J\Phi\}$ and $\{\Psi_1, \Psi_2, \Psi_3\}$ are local orthonormal frame of LM and $\wedge_0^{1,1} M$ respectively.

We denote by $\mathcal{W}$ the Weyl conformal curvature operator of M and $\mathcal{W}^+$ (resp. $\mathcal{W}^-$) the so called self-dual (resp. anti-self-dual) Weyl curvature operator. Considering the operators $\mathcal{R}$ and $\mathcal{W}$ as symmetric endomorphisms of $\Lambda^2 M$, we have the following $SO(4)$ -splitting

$$\mathcal{R} = \frac{\tau}{12} Id + \widetilde{\text{Ric}}_0 + \mathcal{W}^+ + \mathcal{W}^-,$$

where $\widetilde{\text{Ric}}_0$ is the Kulkarni-Nomizu extension of the traceless Ricci tensor Ric_0 to an endomorphism of $\Lambda^2 M$ anticommuting with the Hodge $*$ -operator. It is well-known that M is an Einstein manifold if and only if the bundles $\Lambda_{\pm}^2 M$ are preserved by the operator $\mathcal{R}$. Further, $\mathcal{W}^+$ decomposes into three pieces corresponding to the decomposition (2.3). With respect to the basis $\{\Omega/\sqrt{2}, \Phi, J\Phi\}$ of $\Lambda_+^2 M$, we can write $\mathcal{W}^+$ as the matrix of the form

$$(2.5) \quad \mathcal{W}^+ = \left(\begin{array}{c|c} \kappa/6 & \mathcal{W}_2^+ \\ \hline (\mathcal{W}_2^+)^* & \mathcal{W}_3^+ - (\kappa/12)I \end{array} \right),$$

where the smooth function $\kappa = 3g(\mathcal{W}^+(\Omega), \Omega)$ is the so called conformal scalar curvature, $\mathcal{W}_2^+$ is the part of $\mathcal{W}^+$ which exchanges the two factors in (2.3) and $\mathcal{W}_3^+$ is a traceless self-adjoint endomorphism of LM ([1]). We may easily check that

$$\kappa = \frac{1}{2}(3\tau^* - \tau) \quad \text{and} \quad \frac{\tau^*}{2} = g(\mathcal{R}(\Omega), \Omega).$$

An almost Hermitian manifold is called a manifold with Hermitian Weyl tensor if it satisfies the condition $\mathcal{W}_3^+ = 0$.

Now, we assume that M is a four-dimensional almost Kähler manifold. Then, the equality

$$R_{ijkl} - R_{ij\bar{k}\bar{l}} - R_{i\bar{j}kl} + R_{i\bar{j}\bar{k}\bar{l}} + R_{i\bar{j}k\bar{l}} + R_{i\bar{j}\bar{k}l} + R_{i\bar{j}kl} + R_{i\bar{j}\bar{k}\bar{l}} = 2 \sum_a (\nabla_a J_{ij})(\nabla_a J_{kl})$$

holds ([3]), and hence we have

$$\|\nabla J\|^2 = 2(\tau^* - \tau).$$

Since almost Kähler manifold is quasi-Kähler, we may observe that there exists a local 1-form α such that

$$\nabla \Omega = \alpha \otimes \Phi - J\alpha \otimes J\Phi.$$

From this equality, we have

$$\|\alpha\|^2 = \frac{1}{2} \|\nabla \Omega\|^2 = \frac{\tau^* - \tau}{2}.$$

We put

$$\begin{aligned} u &= g(\mathcal{R}(\Phi), \Phi) = -(R_{1313} - R_{1324}), \\ v &= g(\mathcal{R}(J\Phi), J\Phi) = -(R_{1414} + R_{1423}), \\ w &= g(\mathcal{R}(\Phi), J\Phi) = -(R_{1314} + R_{1323}), \end{aligned}$$

and define a function K on M by

$$K = (u - v)^2 + 4w^2.$$

If M is in addition an Einstein manifold, then we may observe that

$$u + v = -\frac{\tau^* - \tau}{4}$$

and that the matrix $\mathcal{W}_3^+$ in (2.5) is given by

$$\mathcal{W}_3^+ = \begin{pmatrix} u + \frac{\tau^* - \tau}{8} & w \\ w & v + \frac{\tau^* - \tau}{8} \end{pmatrix},$$

and hence we have the following.

Lemma 3 *A four-dimensional almost Kähler Einstein manifold M is a manifold with Hermitian Weyl tensor if and only if $K = 0$.*

In the sequel, we assume that M is a four-dimensional strictly almost Kähler Einstein manifold. Then, the set $M_0 = \{p \in M \mid (\tau^* - \tau)(p) > 0\}$ is a non-empty open submanifold of M . Further, we put $M_1 = \{p \in M_0 \mid G(p) > 0\}$, where G be a function on M defined by

$$G = \sum_{i,j} (\rho_{ij}^* - \rho_{ji}^*)^2.$$

Then, we see that $M_0 - M_1$ has no interior point unless it is empty in case $\tau < 0$ (see [6]).

Lemma 4 ([6], Lemma 4) *For each point $p \in M_1$, there exists a special adapted pair $\{\{e_i\}_{i=1,\dots,4}, \{\alpha, J\alpha\}\}$ defined on a neighborhood U of p . Namely, the unitary frame $\{e_i\}$ and 1-forms $\alpha, J\alpha$ on U satisfy*

$$\begin{aligned}\nabla\Omega &= \alpha \otimes \frac{1}{\sqrt{2}}(e^1 \wedge e^3 - e^2 \wedge e^4) - J\alpha \otimes \frac{1}{\sqrt{2}}(e^1 \wedge e^4 + e^2 \wedge e^3), \\ \alpha &= \alpha_1 e^1, \quad \alpha_1^2 = \|\alpha\|^2 = \frac{\tau^* - \tau}{2}\end{aligned}$$

and

$$\rho_{13}^* = \frac{\sqrt{G}}{4}, \quad \rho_{14}^* = 0.$$

With respect to the special adapted pair, we have the following equalities ([6], (3.9)–(3.12) and (3.19)):

$$(2.6) \quad \Gamma_{113} = \Gamma_{123} = \Gamma_{124} = \Gamma_{223} = \Gamma_{214} = \Gamma_{224} = 0,$$

$$(2.7) \quad \Gamma_{114} = \Gamma_{213} = -\frac{\alpha_1}{\sqrt{2}},$$

$$(2.8) \quad \Gamma_{324} - \Gamma_{313} = 0, \quad \Gamma_{323} + \Gamma_{314} = 0,$$

$$(2.9) \quad \Gamma_{424} - \Gamma_{413} = 0, \quad \Gamma_{423} + \Gamma_{414} = 0,$$

$$(2.10) \quad \Gamma_{324} - \Gamma_{423} = 0, \quad \Gamma_{313} + \Gamma_{414} = 0,$$

$$(2.11) \quad \rho_{13}^* = \frac{\alpha_1}{\sqrt{2}}(\Gamma_{314} - \Gamma_{413}) = -\frac{\alpha_1}{\sqrt{2}}(\Gamma_{323} + \Gamma_{424})$$

$$(2.12) \quad (e_3\tau^*)^2 + (e_4\tau^*)^2 = 4(\tau^* - \tau)K + \frac{(\tau^* - \tau)^3}{4} - 2(\tau^* - \tau)^2(u - v).$$

Further, the first Chern form γ is locally represented as follows ([6], (2.44) with $\rho_{14}^* = 0$):

$$(2.13) \quad 8\pi\gamma = \tau(e^1 \wedge e^2 + e^3 \wedge e^4) + (\tau^* - \tau)e^3 \wedge e^4 + 4\rho_{13}^*(e^1 \wedge e^4 + e^2 \wedge e^3).$$

3. PROOF OF THEOREM 1

We assume that $G > 0$ everywhere on M . In this case, M is a strictly almost Kähler manifold. Further, the 2-form $\nu(X, Y) = \rho^*(X, Y) - \rho^*(Y, X)$ is J -skew-invariant and perpendicular to Ω , and hence a global section of the vector bundle LM . Thus, the Euler class $e(LM)$ of LM vanishes, and hence the first Chern class $c_1(LM, J)$ of the complex line bundle LM must vanish.

Now, we calculate the first Chern form $\gamma(LM, J)$ of complex line bundle LM . Let $\{\{e_i\}_{i=1,2,3,4}, \{\alpha, J\alpha\}\}$ be a special adapted pair and put

$$\theta^1 = \frac{1}{\sqrt{2}}(e^1 \wedge e^3 - e^2 \wedge e^4), \quad \theta^2 = \frac{1}{\sqrt{2}}(e^1 \wedge e^4 + e^2 \wedge e^3).$$

Then, $\{\theta^1, \theta^2\}$ is a local orthonormal frame field of LM . By direct calculation, we have

$$\begin{aligned} \nabla_{e_1} \theta^1 &= \frac{1}{\sqrt{2}}((\nabla_{e_1} e^1) \wedge e^3 + e^1 \wedge \nabla_{e_1} e^3 - (\nabla_{e_1} e^2) \wedge e^4 - e^2 \wedge \nabla_{e_1} e^4) \\ &= (\Gamma_{112} + \Gamma_{134})\theta^2. \end{aligned}$$

Similarly, we obtain

$$(3.1) \quad \nabla_{e_i} \theta^1 = \Gamma_i \theta^2, \quad \nabla_{e_i} \theta^2 = -\Gamma_i \theta^1,$$

for $i = 1, \dots, 4$, where $\Gamma_i = \Gamma_{i12} + \Gamma_{i34}$. Thus, taking account of (2.6)–(2.11), we have

$$\begin{aligned} (3.2) \quad \nabla_{[e_1, e_2]} \theta^1 &= \sum_a (\Gamma_{12a} - \Gamma_{21a}) \nabla_{e_a} \theta^1 \\ &= \Gamma_{121} \Gamma_1 \theta^2 - \Gamma_{212} \Gamma_2 \theta^2 + \frac{\alpha_1}{\sqrt{2}} \Gamma_3 \theta^2, \\ \nabla_{[e_1, e_3]} \theta^1 &= -\Gamma_{312} \Gamma_2 \theta^2 - \Gamma_{313} \Gamma_3 \theta^2 + (\Gamma_{134} - \Gamma_{314}) \Gamma_4 \theta^2, \\ \nabla_{[e_1, e_4]} \theta^1 &= \frac{\alpha_1}{\sqrt{2}} \Gamma_1 \theta^2 - \Gamma_{412} \Gamma_2 \theta^2 - (\Gamma_{134} + \Gamma_{413}) \Gamma_3 \theta^2 + \Gamma_{313} \Gamma_4 \theta^2, \\ \nabla_{[e_2, e_3]} \theta^1 &= \left(\frac{\alpha_1}{\sqrt{2}} + \Gamma_{312} \right) \Gamma_1 \theta^2 + \Gamma_{314} \Gamma_3 \theta^2 + (\Gamma_{234} - \Gamma_{313}) \Gamma_4 \theta^2, \\ \nabla_{[e_2, e_4]} \theta^1 &= \Gamma_{412} \Gamma_1 \theta^2 - (\Gamma_{234} + \Gamma_{313}) \Gamma_3 \theta^2 - \Gamma_{413} \Gamma_4 \theta^2, \\ \nabla_{[e_3, e_4]} \theta^1 &= -\frac{\sqrt{2}}{\alpha_1} \rho_{13}^* \Gamma_1 \theta^2 + \Gamma_{343} \Gamma_3 \theta^2 - \Gamma_{434} \Gamma_4 \theta^2. \end{aligned}$$

We denote by K the curvature tensor of LM . Then, taking account of (3.1), (3.2), (2.6)–(2.11), we have

$$\begin{aligned} K(e_1, e_2) \theta^1 &= \nabla_{e_1} (\nabla_{e_2} \theta^1) - \nabla_{e_2} (\nabla_{e_1} \theta^1) - \nabla_{[e_1, e_2]} \theta^1 \\ &= \nabla_{e_1} (\Gamma_2 \theta^2) - \nabla_{e_2} (\Gamma_1 \theta^2) + \Gamma_{112} \Gamma_1 \theta^2 + \Gamma_{212} \Gamma_2 \theta^2 - \frac{\alpha_1}{\sqrt{2}} \Gamma_3 \theta^2 \\ &= (e_1 \Gamma_{212} + e_1 \Gamma_{234}) \theta^2 - (e_2 \Gamma_{112} + e_2 \Gamma_{134}) \theta^2 \end{aligned}$$

$$\begin{aligned}
& +\Gamma_{112}(\Gamma_{112} + \Gamma_{134})\theta^2 + \Gamma_{212}(\Gamma_{212} + \Gamma_{234})\theta^2 \\
& -\frac{\alpha_1}{\sqrt{2}}(\Gamma_{312} + \Gamma_{334})\theta^2 \\
= & \left(R_{1212} + R_{1234} + \frac{\tau^* - \tau}{4} \right) \theta^2 = \left(-\rho_{11}^* + \frac{\tau^* - \tau}{4} \right) \theta^2 \\
= & -\frac{\tau}{4}\theta^2.
\end{aligned}$$

Similarly, we obtain

$$\begin{aligned}
K(e_1, e_3)\theta^1 &= \rho_{14}^*\theta^2 = 0, \\
K(e_1, e_4)\theta^1 &= -\rho_{13}^*\theta^2, \\
K(e_2, e_3)\theta^1 &= \rho_{24}^*\theta^2 = -\rho_{13}^*\theta^2, \\
K(e_2, e_4)\theta^1 &= -\rho_{23}^*\theta^2 = 0, \\
K(e_3, e_4)\theta^1 &= -\rho_{33}^*\theta^2 = -\frac{\tau^*}{4}\theta^2.
\end{aligned}$$

Hence, we have

$$\begin{aligned}
2\pi\gamma(LM, J) &= -\left(-\frac{\tau}{4}e^1 \wedge e^2 - \rho_{13}^*e^1 \wedge e^4 - \rho_{13}^*e^2 \wedge e^3 - \frac{\tau^*}{4}e^3 \wedge e^4 \right) \\
&= \frac{1}{4}\{\tau(e^1 \wedge e^2 + e^3 \wedge e^4) + (\tau^* - \tau)e^3 \wedge e^4 \\
&\quad + 4\rho_{13}^*(e^1 \wedge e^4 + e^2 \wedge e^3)\}.
\end{aligned}$$

Comparing (2.13) with this equality, we conclude that

$$(3.3) \quad c_1(M) = c_1(LM, J) = 0.$$

We now recall the Wu's theorem and the equality established in [9, (2.9)]:

$$(3.4) \quad p_1(M) + 2\chi(M) = c_1(M)^2,$$

$$(3.5) \quad p_1(M) + 2\chi(M) = \frac{1}{2\pi^2} \int_M \|\mathcal{R}_+\|^2 dM,$$

where $p_1(M)$ and $\chi(M)$ are the Pontrjagin class and the Euler-Poincaré class respectively, and $\mathcal{R}_+$ is the restriction of $\mathcal{R}$ to $\wedge_+^2 M$. From (3.3), (3.4) and (3.5), we have $\mathcal{R}_+ = 0$ and hence $u = v = w = 0$, $\tau^* = 0$. Therefore, M is a Ricci flat Kähler manifold. But, this is a contradiction and completes the proof.

4. ANOTHER PROOF OF THEOREM 2

We assume that M is not a Kähler manifold. Then, the submanifold M_0 is not empty. Let $p_0 \in M_0$ be a point at which $\tau^* - \tau$ takes its maximum value.

We suppose $p_0 \in M_1$. Then, from Lemma 3 and (2.12), we obtain $(\tau^* - \tau)(p_0) = 0$. But, this is a contradiction.

Next, we suppose $p_0 \notin M_1$, namely $G(p_0) = 0$. If $M_1^c = M - M_1$ has an interior point, then from the result in [7], it follows that M is a Ricci flat Kähler manifold. But, this is also a contradiction. Thus, M_1^c has no interior point. Let $\{p_n\}_{n=1,2,\dots}$ be a sequence in M_1 with $p_n \rightarrow p_0$ as $n \rightarrow \infty$. Then, from (2.12), we get

$$\|\text{grad}(\tau^* - \tau)\|^2(p_n) \geq (e_3\tau^*)^2(p_n) + (e_4\tau^*)^2(p_n) = \frac{(\tau^* - \tau)^3(p_n)}{4} (> 0)$$

for all n . On one hand, from the continuity the function $\|\text{grad}(\tau^* - \tau)\|^2$, we have

$$\lim_{n \rightarrow \infty} \|\text{grad}(\tau^* - \tau)\|^2(p_n) = \|\text{grad}(\tau^* - \tau)\|^2(p_0) = 0.$$

Therefore, we obtain $(\tau^* - \tau)(p_0) = 0$, which is a contradiction. This completes the proof.

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ON A CLASS OF LOCALLY CONFORMAL MANIFOLDS

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ABSTRACT. Geometrical and structural properties are proved for locally conformal almost cosymplectic manifolds.

1. INTRODUCTION

Let M be a $(2m + 1)$ -dimensional Riemannian manifold. We assume that M is structured to be a locally conformal almost cosymplectic manifold $M(\Omega, \phi, \eta, \xi, f)$, where

- (i) Ω is a structure 2-form of rank $2m$,
- (ii) ϕ is a tensor field of type (1.1) and of rank $2m$ which satisfies

$$\phi^2 = -\text{Id} + \eta \otimes \xi,$$

- (iii) η is the structure 1-form,
- (iv) ξ is the structure vector field (Reeb vector field),
- (v) f is a nonzero constant.

If $d^\omega = d + e(\omega)$ means the cohomological operator, one has

$$d^{2f\eta}\Omega = 0, \quad (\text{or equivalently } d\Omega = -2f\eta \wedge \Omega).$$

The vector field $V(V^a)(a, b \in \{1, \dots, 2m\})$ such that

$$(1) \quad dV^a + V^b\theta_b^a = V^a\eta, \quad \eta(V) = C \quad (C = \text{const.}),$$

(where θ_b^a are the local connection forms in the tangent bundle) is a principal vector field on M . Taking the Lie derivative of Ω with respect to V , one derives

$$(2) \quad \mathcal{L}_V \Omega = -2fC\Omega + 2(1+f)\eta \wedge \beta,$$

where

$$(3) \quad \beta = \sum (V^i \omega^{i^*} - V^{i^*} \omega^i), \quad (i = 1, \dots, m, i^* = i + m),$$

where the ω^a form the associated coframe of the adapted vectorial frame $\mathcal{O} = \text{vect}\{e_a, \xi\}$. By exterior differentiation one gets

$$(4) \quad d(\mathcal{L}_V \Omega) = \eta \wedge \mathcal{L}_V \Omega,$$

which shows that $\mathcal{L}_V \Omega$ is an exterior recurrent form [8] having the structure form η as recurrence form. Next, one has

$$(5) \quad \nabla(\phi V) = \phi V \wedge \xi,$$

(where $\wedge$ stands for the wedge product of vector fields), which proves that ϕV is a Killing vector field having the Reeb vector field ξ as generative [10]. On the other hand, if $[,]$ denotes the Lie bracket, one derives

$$(6) \quad [V, \xi] = 0, \quad [\phi V, \xi] = 0, \quad [V, \phi V] = 0,$$

which means that $V, \phi V$, and ξ define a commutative triple [11] [6] [2].

Setting $\|V\|^2 = l$, one derives

$$(7) \quad dl = l\eta, \quad \text{grad } l = l\xi, \quad \text{div}(\text{grad } l) = 2ml, \quad \|\text{grad } l\|^2 = l^2.$$

From (7) it is seen that $\|\text{grad } l\|^2$ and $\text{div}(\text{grad } l)$ are functions of l ; following [12], this proves that l is an isoparametric function.

Some further properties following from (7) are also discussed.

2. PRELIMINARIES

Let (M, g) be a Riemannian C^∞ -manifold and let ∇ be the covariant differential operator defined with respect to the metric tensor g . We assume in the sequel that M is oriented and that ∇ is the Levi-Civita connection. Let $\Gamma TM = \Xi(M)$ be the set of sections of the tangent bundle TM , and

$$\flat : TM \xrightarrow{\flat} T^*M \quad \text{and} \quad \sharp : TM \xleftarrow{\sharp} T^*M$$

the classical isomorphisms defined by g (i.e. $\flat$ is the index lowering operator, and $\sharp$ is the index raising operator).

Following [8], we denote by

$$A^q(M, TM) = \Gamma\text{Hom}(\Lambda^q TM, TM),$$

the set of vector valued q -forms ($q < \dim M$), and we write for the covariant derivative operator with respect to ∇

$$d^\nabla : A^q(M, TM) \rightarrow A^{q+1}(M, TM).$$

It should be noticed that in general $d^{\nabla^2} = d^\nabla \circ d^\nabla \neq 0$, unlike $d^2 = d \circ d = 0$. We denote by $dp \in A^1(M, TM)$ the canonical vector valued 1-form of M , which is also called the soldering form of M [3]. Since ∇ is symmetric one has that $d^\nabla(dp) = 0$.

The operator

$$(8) \quad d^\omega = d + e(\omega),$$

acting on ΛM is called the cohomology operator [5]. In (8) $e(\omega)$ means the exterior product by the closed 1-form ω , i.e.

$$(9) \quad d^\omega u = du + \omega \wedge u.$$

For any $u \in \Lambda M$ one has

$$(10) \quad d^\omega \circ d^\omega = 0.$$

A form u such that $d^\omega u = 0$ is said to be d^ω -closed.

A function $\epsilon : \mathbb{R}^n \rightarrow \mathbb{R}$ is isoparametric [12] if $\|\nabla\epsilon\|^2$ and $\text{div}(\nabla\epsilon)$ are functions of ϵ ($\nabla\epsilon = \text{grad}\epsilon$).

Let $\mathcal{O} = \text{vect}\{e_A | A = 1, \dots, 2m+1\}$ be a local field of adapted vectorial frames over M and let $\mathcal{O}^* = \text{covect}\{\omega^A\}$ be its associated coframe. Then the soldering form dp is expressed by

$$(11) \quad dp = \omega^A \otimes e_A,$$

and E. Cartan's structure equations can be written in indexless manner are

$$(12) \quad \nabla e = \theta \otimes e,$$

$$(13) \quad d\omega = -\theta \wedge \omega,$$

$$(14) \quad d\theta = -\theta \wedge \theta + \Theta.$$

In the above equations θ (respectively Θ) are the local connection forms in the tangent bundle TM (respectively the curvature 2-forms on M).

3. LOCALLY CONFORMAL MANIFOLDS

Let $M(\Omega, \phi, \eta, \xi, f)$ be a locally conformal almost cosymplectic manifold as defined in Section 1, i.e.

$$(15) \quad d^2 f \eta \Omega = 0.$$

Consider the vector field

$$(16) \quad V = \sum V^A e_A, \quad A = 0, 1, \dots, m,$$

and assume that V is a pfaffian transformation, i.e.

$$(17) \quad \mathcal{L}_V \eta = 0 \iff \eta(V) = C = V^0 = \text{constant}.$$

Taking the covariant differential of V , and making use of (12) and (17), one finds

$$(18) \quad \nabla V = (dV^a + V^b \theta_b^a) \otimes e_a + V^0 dp - V^b \otimes \xi.$$

We assume in the sequel that the following relation

$$(19) \quad dV^a + V^b \theta_b^a = V^a \eta$$

holds. Under this condition, we agree to call V the principal vector field on M . Under these assumptions, and since in general one has from the structure equations

$$(20) \quad \nabla \xi = dp - \eta \otimes \xi,$$

one derives

$$(21) \quad \nabla V = C dp + V \wedge \xi - C \eta \otimes \xi, \quad \text{where } C = V^0,$$

and where $\wedge$ denotes the wedge product of vector fields.

Making use of the operator ϕ (see Section 1), one also derives that

$$(22) \quad \nabla(\phi V) = \phi V \wedge \xi.$$

This proves the fact that ϕV is a Killing vector field having ξ as generative. By (20), (21), and (22), one may write

$$(23) \quad [V, \xi] = 0, \quad [\phi V, \xi] = 0, \quad [V, \phi V] = 0,$$

where $[\ , \]$ means the Lie bracket. This proves that $V, \phi V$, and ξ define a commutative triple [11] [6] [2].

On the other hand, since

$$(24) \quad V^b = \sum_a V^a \omega^a + V^0 \eta,$$

one derives by (19) and (13) that

$$(25) \quad dV^b = \eta \wedge V^b.$$

This expresses that the dual form V^b of the principal vector field V is exterior recurrent [1] with the same recurrent 1-form as $\mathcal{L}_V \Omega$ (see (4)). Taking now the Lie derivative of Ω with respect to V , one gets

$$(26) \quad \mathcal{L}_V \Omega = -2fC\Omega + 2f\eta \wedge \beta + d\beta,$$

where we have set

$$(27) \quad \beta = \sum (V^i \omega^{i*} - V^{i*} \omega^i), \quad i = 1, \dots, m; i^* = i + m.$$

Further, by differentiation of β , one gets by (19) and the structure equations (13) that

$$(28) \quad d\beta = V^i (\omega^a \wedge \theta_a^{i*} + \eta \wedge \omega^{i*}) - V^{i*} (\omega^a \wedge \theta_a^i + \eta \wedge \omega^i).$$

Now, using the Kaehlerian relations

$$(29) \quad \theta_j^i = \theta_{j^*}^{i*}, \quad \theta_j^{i*} = \theta_i^{j*},$$

there finally follows that

$$(30) \quad d\beta = 2\eta \wedge \beta.$$

Hence, the above equation says that the 1-form β is exterior recurrent with (up to a factor 2) the structure 1-form η as recurrent 1-form.

Under the above conditions, it follows by (30) that (26) that

$$(31) \quad \mathcal{L}_V \Omega = -2fC\Omega + 2(f+1)\eta \wedge \beta,$$

and differentiation of (31) gives

$$(32) \quad d(\mathcal{L}_V \Omega) = \eta \wedge \mathcal{L}_V \Omega.$$

Hence, as in the case of (30), one may say that the Lie derivative $\mathcal{L}_V \Omega$ is exterior recurrent with η as recurrence form. It should be noticed that (32) can also be expressed as

$$(33) \quad d^{-\eta}(\mathcal{L}_V \Omega) = 0,$$

and that $\mathcal{L}_V \omega$ is consequently $d^{-\eta}$ -closed (8).

Since the q -th covariant differential ∇^q for a vector field Z on M is defined inductively as

$$\nabla^q Z = d^\nabla (\nabla^{q-1}) Z,$$

one finds in the case under consideration

$$(34) \quad \nabla^2 V = V^\flat \otimes dp + (\eta \wedge V^\flat) \otimes \xi,$$

and

$$(35) \quad \nabla^3 V = 0.$$

By reference to [7], one may hence say that V is exterior quasi concurrent and a zero element of $A^3(M, TM)$ (see Section 1).

Let $\mathbb{L}$ be the operator of type (1.1) on forms as defined by S. Goldberg [4], i.e.

$$(36) \quad \mathbb{L}u = u \wedge \Omega, \quad u \in \Lambda^1 M.$$

Setting $u = V^\flat$ one finds by (25)

$$(37) \quad d(\mathbb{L}u) = (1 - 2f)\eta \wedge \mathbb{L}u,$$

and

$$(38) \quad d(\mathbb{L}^q u) = (1 - 2f)\eta \wedge \mathbb{L}^q u.$$

From the above formulas it can be noticed that all $(2q + 1)$ -forms $\mathbb{L}^q u$ are exterior recurrent with the closed form $(1 - 2f)\eta$ as recurrence form (we recall that $1 - 2f$ is a constant).

Invoking (19), one now calculates that

$$(39) \quad d\|V\|^2 = \|V\|^2 \eta,$$

and consecutively by (25) one gets

$$(40) \quad \mathcal{L}_V V^b = C V^b.$$

Following [10] it follows from (40) that V is a selfconformal vector field. By reference to (21) one now remarks that the condition for V to be an affine vector field or a Killing vector field are mutually equivalent (that is $C = 0$), i.e.

$$\mathcal{L}_V V^b = 0, \quad \nabla V = V \wedge \xi.$$

Setting $\|V\|^2 = l$, it follows by (39) that

$$(41) \quad \text{grad } l = l\xi,$$

and so one gets

$$\text{div}(\text{grad } l) = 2ml, \quad \|\text{grad } l\|^2 = l^2.$$

Hence, since $\text{div}(\text{grad } l)$ and $\|\text{grad } l\|^2$ are functions of l , then following [12], $l : \mathbb{R} \rightarrow \mathbb{R}$ is an isoparametric function.

Continuing along this line of ideas, and operating on (41) with the operator ∇ and taking into account (20), one finds that

$$(42) \quad \nabla \text{grad } l = l \, dp.$$

Then, if $Z, Z' \in \Xi(M)$ are any vector fields, one gets from (42)

$$(43) \quad \langle \nabla_Z \text{grad } l, Z' \rangle = l \langle Z, Z' \rangle,$$

which shows that $\text{grad } l$ defines an infinitesimal concircular transformation [13].

Finally, operating on (42) by d^∇ , one gets

$$(44) \quad \nabla^2 \text{grad } l = l\eta \wedge dp.$$

This proves that $\text{grad } l$ is an exterior concurrent vector field [9] [7], and consequently one has

$$(45) \quad \mathcal{R}(\text{grad } l, Z) = 2ml \, g(\text{grad } l, Z), \quad Z \in \Xi(M),$$

where $\mathcal{R}$ means the Ricci curvature.

Summarizing, we can state the following

Theorem 3.1 *Let $M(\Omega, \phi, \eta, \xi, f)$ be a locally conformal almost cosymplectic manifold and V be the principal vector field on M ($\eta(V) = C = \text{const.}$). One has the following properties:*

(i) *the Lie derivative of the structure 2-form Ω with respect to V , is exterior recurrent with the structure 1-form η as recurrence form, i.e.*

$$d(\mathcal{L}_V \Omega) = \eta \wedge \mathcal{L}_V \Omega;$$

(ii) *the vector field ϕV is a Killing vector field with the Reeb vector field $\xi = \eta^\sharp$ as generative;*

(iii) *$V, \phi V$, and ξ define a commutative triple, i.e.*

$$[V, \xi] = 0, \quad [\phi V, \xi] = 0, \quad [V, \phi V] = 0,$$

where $[,]$ denotes the Lie bracket;

(iv) *Setting $\|V\|^2 = l$, one derives that*

$$\text{div}(\text{grad } l) = 2ml, \quad \|\text{grad } l\|^2 = l^2,$$

which proves that, according to [12], l is an isoparametric function;

(v) *if $\mathbb{L}$ is the operator of S . Goldberg, and setting $u = V^\flat$, then all $(2q+1)$ -forms $\mathbb{L}^q u$ are exterior recurrent, having up to a constant η as recurrence form;*

(vi) *V is a self conformal vector field, i.e.*

$$\mathcal{L}_V V^\flat = C V^\flat, \quad C = \text{const.};$$

(vii) *$\text{grad } l$ is an exterior concurrent vector field, i.e.*

$$\nabla^2 \text{grad } l = l\eta \wedge dp \Rightarrow \mathcal{R}(\text{grad } l, Z) = 2ml \, g(\text{grad } l, Z), \quad Z \in \Xi(M).$$

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SEMITOPOLOGICAL SPACES

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ABSTRACT. Levine (1963) introduced the notions of semi-open set and semi-continuity in topological spaces and obtained a number of their properties. Biswas (1969) defined semi-open functions and investigated several properties of such functions. Also, Noiri (1973) investigated some properties of semi-continuity and semi-open functions. Latif (1998) introduced the concept of semitopology and explored some of its properties. In this paper we give a characterization of S -semi-continuous functions, S -irresolute functions, separation axioms, S -semi-closed graphs and strongly S -semi-closed graphs. It is shown that many results in previous papers can be considered as special cases of our results.

1. INTRODUCTION

Let X , Y , and Z be topological spaces on which no separation axioms are assumed unless explicitly stated. Let S be a subset of X . The closure (resp. interior, boundary) of S will be denoted by $CI(S)$ (resp. $Int(S)$, $Bd(S)$). A subset S of X is called semi-open (Levine 1963) if $S \subseteq CI[Int(S)]$, the complement of a semi-open set is called semi-closed. The family of all semi-open sets in X will be denoted by $SO(X)$. The semi-interior of a set A is the largest semi-open set contained in A . A function $f : X \rightarrow Y$ is called semi-continuous (Levine 1963) if $f^{-1}(V) \in SO(X)$ for each open set V of Y , f is irresolute (Crossley and Hildebrand 1971) if $f^{-1}(V) \in SO(X)$ for each $V \in SO(Y)$. A function $f : X \rightarrow Y$ is called semi-open (Biswas 1969) if $f(U) \in SO(Y)$ for each open set U of X , and it is called semi-closed (Noiri 1973) if the image of each closed set in X is semi-closed in Y .

2. SEMITOPOLOGIES

The following definitions are taken from our earlier papers of 1998.

Definition 1.1 Let (X, τ) be a topological space. Let $\gamma \subseteq SO(X, \tau)$. Then we define the semi-intersection of members of γ to be the semi-interior of $\cap \gamma$, denoted by $\cap^* \gamma$. If $A_1, A_2, \dots, A_n \in SO(X, \tau)$, we will write $\cap^* \{A_1, A_2, \dots, A_n\} = A_1 \cap^* A_2 \cap^* \dots \cap^* A_n$.

Definition 1.2 Let (X, τ) be a topological space. Let $\tau^* \subseteq SO(X, \tau)$. We say that τ^* is a semitopology on X if the following axioms are satisfied:

- (S₁) $X, \phi \in \tau^*$,
- (S₂) For every $A, B \in \tau^*$, $A \cap^* B \in \tau^*$,
- (S₃) For each $\gamma \subseteq \tau^*$, $\cup \gamma \in \tau^*$.

We will call the ordered pair (X, τ^*) a semitopological space.

Definition 1.3 Let (X, τ) be a topological space and τ^* be a semitopology on X . A semi-open set A in X will be called S -semi-open in X if and only if $A \in \tau^*$.

Definition 1.4 Given a semi-topological space (X, τ^*) , a semi-open subset E of X is said to be S -semi-closed in X when its complement $X - E$ in X is S -semi-open in X .

Various notations like the interior, closure, exterior and the derived set operators as well as set properties like dense, nowhere dense-in itself, compact etc. can be defined in a semitopological space in analogy with topological spaces. Many results of topological spaces remain valid in semitopological spaces, whereas some become false. The S -semi-derived set (resp. S -semi-closure, S -semi-interior, S -semi-boundary) of a subset S of a space X will be denoted by $sDer(S)$ (resp. $sCl(S)$, $sInt(S)$, $sBd(S)$).

In what follows, we list the main properties of such operations which give the deviations between these operations and that in topological spaces.

Theorem 1.1 *Let (X, τ) be a topological space and let $\tau^* \subseteq SO(X, \tau)$ such that τ^* is a semitopology on X . Let A and B be subsets of X . Then*

- (i) $sDer(A) \cup sDer(B) \subseteq sDer(A \cup B)$;
- (ii) $sCI(A) \cup sCI(B) \subseteq sCI(A \cup B)$;
- (iii) $sInt(A \cap B) \subseteq sInt(A) \cap sInt(B)$.

It can be easily shown, by examples, that the inequalities in (i), (ii) and (iii) cannot be replaced, in general, by equalities as in the case of topological spaces.

Definition 2.2 Let (X, τ) be a topological space and $\tau^* \subseteq SO(X, \tau)$ such that τ^* is a semitopology on X . We call τ^* a semitopology associated with τ if $\tau \subseteq \tau^*$.

Example. Let (X, τ) be any topological space. Then $\tau^* = SO(X, \tau)$ is a semitopology associated with τ .

Remark. The intersection of two S -semi-open sets need not be S -semi-open. But if $U \in \tau$ and $V \in \tau^*$, then $U \cap V \in \tau^*$.

2. S -SEMI-CONTINUOUS FUNCTIONS

Definition 2.1 Let (X, τ_1) (Y, τ_2) be topological spaces and τ_1^* be an associated semitopology with τ_1 . We say that a function $f : X \rightarrow Y$ is an S -semi-continuous function if the inverse image of each open set in Y is S -semi-open in X .

Theorem 2.1 Let $f : (X, \tau_1) \rightarrow (Y, \tau_2)$ be a function. Then the following are equivalent:

- (i) f is S -semi-continuous;
- (ii) The inverse image of each closed set in Y is S -semi-closed in X ;
- (iii) $sCI[f^{-1}(V)] \subseteq f^{-1}[Cl(V)]$, for every $V \subseteq Y$;
- (iv) $f[sCI(U)] \subseteq Cl[f(U)]$, for every $U \subseteq X$;
- (v) For any point $x \in X$ and any open set V of Y containing $f(x)$, there exists $U \in \tau_1^*$ such that $x \in U$ and $f(U) \subseteq V$;
- (vi) $sBd[f^{-1}(V)] \subseteq f^{-1}[Bd(V)]$, for every $V \subseteq Y$;
- (vii) $f[sCl(U)] \subseteq Cl[f(U)]$, for every $U \subseteq X$;
- (viii) $f^{-1}(Int(B)) \subseteq sInt[f^{-1}(B)]$, for every $B \subseteq Y$;

Proof. (i) $\Rightarrow$ (ii). Let $F \subseteq Y$ be closed. Since f is S -semi-continuous, $f^{-1}(Y - F) = X - f^{-1}(F)$ is S -semi-open. Therefore, $f^{-1}(F)$ is S -semi-closed in X .

(ii) $\Rightarrow$ (iii). Since $Cl(V)$ is closed for every $V \subseteq Y$, then $f^{-1}[Cl(V)]$ is S -semi-closed. Therefore $f^{-1}[Cl(V)] = sCl[f^{-1}(Cl(V))] \supseteq sCl[f^{-1}(V)]$.

(iii) $\Rightarrow$ (iv). Let $U \subseteq X$ and $f(U) = V$. Then $f^{-1}[Cl(V)] \supseteq sCl[f^{-1}(V)]$. Thus $f^{-1}[Cl(f(U))] \supseteq sCl[f^{-1}(f(U))] \supseteq sCl(U)$ and $Cl[f(U)] \supseteq f[sCl(U)]$.

(iv) $\Rightarrow$ (ii). Let $W \subseteq Y$ be a closed set, $U = f^{-1}(W)$, then $f[sCl(U)] \subseteq Cl[f(U)] = Cl[f(f^{-1}(W))] \subseteq Cl(W) = W$. $sCl(U) \subseteq f^{-1}[f(sCl(U))] \subseteq f^{-1}(W) = U$, $sCl(U) \subseteq U$. So U is S -semi-closed.

(ii) $\Rightarrow$ (i). Let $V \subseteq Y$ be an open set. Then $Y - V$ is closed. Therefore $f^{-1}(Y - V) = X - f^{-1}(V)$ is S -semi-closed in X and hence $f^{-1}(V)$ is S -semi-open in X .

(i) $\Rightarrow$ (v). Let $f : X \rightarrow Y$ be S -semi-continuous. For any $x \in X$ and any open set V of Y containing $f(x)$, $U = f^{-1}(V) \in \tau_1^*$ and $f(U) = f[f^{-1}(V)] \subseteq V$.

(v) $\Rightarrow$ (i). Let $V \in \tau_2$. We prove that $f^{-1}(V) \in \tau_1^*$. Let $x \in f^{-1}(V)$. Then $f(x) \in (V)$ and there exists $U \in \tau_1^*$ such that $x \in U$ and $f(x) \in f(u) \subseteq V$. Hence $x \in U \subseteq f^{-1}[f(U)] \subseteq f^{-1}(V)$ and $f^{-1}(V)$ is S -semi-open neighborhood of each of its points. Therefore $f^{-1}(V) \in \tau_1^*$.

(i) $\Rightarrow$ (vi). Since $f^{-1}[Bd(V)] = f^{-1}[Cl(V) - Int(V)]$. For each $V \subseteq Y$, $f^{-1}[Bd(V)] = f^{-1}[Cl(V)] - f^{-1}[Int(V)] \supseteq sCl[f^{-1}(V)] - sInt[f^{-1}(Int(V))] \supseteq sCl[f^{-1}(V)] - sInt[f^{-1}(V)] = sBd[f^{-1}(V)]$.

(vi) $\Rightarrow$ (i). Let $U \subseteq Y$ be an open set, $V = Y - U$. Since V is closed, $sBd[f^{-1}(V)] \subseteq f^{-1}[Bd(V)] \subseteq f^{-1}[Cl(V)] = f^{-1}(V)$. Thus $f^{-1}(V)$ is S -semi-closed and f is S -semi-continuous.

(i) $\Rightarrow$ (vii). It is obvious, since f is S -semi-continuous and by (iv) $f[sCl(U)] \subseteq Cl[f(U)]$ for each $U \subseteq X$. So $f[sDer((U))] \subseteq Cl[f(U)]$.

(vii) $\Rightarrow$ (i). Let $U \subseteq Y$ be an open set, $V = Y - U$ and $f^{-1}(V) = W$. Then $f[sDer(W)] \subseteq Cl[f(W)]$. Thus $f[sDer(f^{-1}(V))] \subseteq Cl[f(f^{-1}(V))] \subseteq Cl(V) = V$. Then $sDer[f^{-1}(V)] \subseteq f^{-1}(V)$ and $f^{-1}(V)$ is S -semi-closed.

Therefore, f is S -semi-continuous.

(i) $\Rightarrow$ (viii). Let $B \subseteq Y$. Then $f^{-1}[Int(B)]$ is S -semi-open in X . But $f^{-1}[Int(B)] = sInt[f^{-1}(Int(B))] \subseteq sInt[f^{-1}(B)]$. Therefore, $f^{-1}[Int(B)] \subseteq sInt[f^{-1}(B)]$.

(viii) $\Rightarrow$ (i). Let $V \subseteq Y$ be an open set. Then $f^{-1}(V) = f^{-1}[Int(V)] \subseteq sInt[f^{-1}(V)]$. Therefore $f^{-1}(V)$ is S -semi-open. Hence f is S -semi-continuous.

Remarks

(i) Every continuous function is S -semi-continuous.

(ii) If a function $f : (X, \tau_1) \rightarrow (Y, \tau_2)$ is S -semi-continuous and a function $g : (Y, \tau_2) \rightarrow (Z, \tau_3)$ is S -semi-continuous, then $g \circ f : (X, \tau_1) \rightarrow (Z, \tau_3)$ may not be S -semi-continuous.

(iii) If a function $f : (X, \tau_1) \rightarrow (Y, \tau_2)$ is S -semi-continuous and $g : (Y, \tau_2) \rightarrow (Z, \tau_3)$ is continuous, then $(g \circ f)$ is S -semi-continuous.

(iv) Let (X, τ_1) and (Y, τ_2) be topological spaces, τ_1^* and τ_2^* be two associated semitopologies with τ_1 and τ_2 , respectively. If $f : X \rightarrow Y$ is a function, and one of the following properties holds

(a) $f^{-1}[sInt(B)] \subseteq Int[f^{-1}(B)]$, for each $B \subseteq Y$,

(b) $Cl[f^{-1}(B)] \subseteq f^{-1}[sCl(B)]$, for each $B \subseteq Y$,

(c) $f[Cl(A)] \subseteq sCl[f(A)]$, for each $A \subseteq X$,

then f is continuous.

Important Remark. Many properties of semi-continuity (Levine 1963) can be easily deduced from the previous results of S -semi-continuity by setting $\tau^* = SO(X, \tau)$.

3. S -IRRESOLUTE FUNCTIONS

Definition 3.1 Let $(X, \tau_1), (Y, \tau_2)$ be topological spaces and τ_1^* and τ_2^* be two associated semitopologies with τ_1, τ_2 , respectively. Let $f : X \rightarrow Y$ be a function. Then we say that f is S -irresolute if the inverse image of each S -semi-open set is S -semi-open.

From the above definition, one can draw the following diagram:

Continuity $\Rightarrow$ S -semi-continuity $\Leftarrow$ S -irresolute

It is easy to give examples to show that the concepts of continuity and S -irresolute are independent.

Theorem 3.1 *Let $f : X \rightarrow Y$ be an injective S -irresolute function. Then $Int[f(A)] \subseteq sInt[f(A)] \subseteq f[sInt(A)]$.*

Theorem 3.2 *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two functions. If f is S -irresolute and g is S -semi-continuous, then $gof : X \rightarrow Z$ is S -irresolute.*

Theorem 3.3 *Let $f : X \rightarrow Y$ be a function. Then the following are equivalent:*

- (i) f is S -irresolute.
- (ii) The inverse image of each S -semi-closed set is S -semi-closed.
- (iii) For each $A \subseteq X$, $f[sCl(A)] \subseteq sCl[f(A)] \subseteq Cl[f(A)]$.
- (iv) For each $B \subseteq Y$, $sCl[f^{-1}(B)] \subseteq f^{-1}[sCl(B)] \subseteq f^{-1}[Cl(B)]$.
- (v) For each $x \in X$ and each S -semi-open set $V \subseteq Y$ containing $f(x)$, there exists an S -semi-open set $U \subseteq X$ containing x such that $f(U) \subseteq V$.
- (vi) For each $B \subseteq Y$, $sBd[f^{-1}(B)] \subseteq f^{-1}[sBd(B)] \subseteq f^{-1}[Bd(B)]$.
- (vii) For each $A \subseteq X$, $f[sDer(A)] \subseteq sCl[f(A)] \subseteq Cl[f(A)]$.
- (viii) For each $B \subseteq Y$, $f^{-1}[Int(B)] \subseteq f^{-1}[sInt(B)] \subseteq sInt[f^{-1}(B)]$.

Proof. Similar to that of Theorem 2.1.

Remarks

- (i) Many properties of irresolute (Crossley and Hildebrand 1971) functions are special cases of previous results of S -irresolute functions by setting $\tau_1^* = SO(X)$ and $\tau_2^* = SO(Y)$.
- (ii) One may also define S -semi-open, S -semi-closed, S -pre-semi-open and S -pre-semi-closed sets as well as S -semi-homeomorphic functions.

4. SEPARATION AXIOMS IN SEMITOPOLOGICAL SPACES

Definition 4.1. Let (X, τ) be a topological space and τ^* be a semitopology associated with τ . Then the semitopological space (X, τ^*) is called

- (i) S -semi- T_0 , if for every two points of X , there exists an S -semi-open neighborhood of one of them to which the other does not belong.
- (ii) S -semi- T_1 , if for every two distinct points x and y in X , there exist two

S -semi-open sets U and V such that $x \in U, y \notin U$ and $y \in V, x \notin V$.

(iii) S -semi- T_2 , if for every two distinct points x and y in X , there exist two disjoint S -semi-open sets U and V such that $x \in U$ and $y \in V$.

(iv) S -semi- T'_2 , if for every two distinct points x and y in X , there exist two S -semi-open sets U and V such that $x \in U, y \in V$ and $Cl(U \cap Cl(V)) = \Phi$.

From the above definitions, one can draw the following diagram:

$$\begin{array}{ccccccc} T'_2 & \Rightarrow & T_2 & \Rightarrow & T_1 & \Rightarrow & T_0 \\ \Downarrow & & \Downarrow & & \Downarrow & & \Downarrow \\ S\text{-semi-}T'_2 & \Rightarrow & S\text{-semi-}T_2 & \Rightarrow & S\text{-semi-}T_1 & \Rightarrow & S\text{-semi-}T_0 \end{array}$$

It is easy to give examples to show that the inverse directions are not true in general.

Theorem 4.1 *Let (X, τ) be a topological space, and τ^* a semitopology on X associated with τ . Suppose the following properties hold:*

- (i) *If $x, y \in X$ such that $x \neq y$, then $SCL(\{x\}) \neq sCl(\{y\})$.*
- (ii) *If $x \in X$, then $sDer(\{x\})$ is the union of S -semi-closed sets.*
- (iii) *If $A \subseteq X$, then $A = \cap\{N : N \neq y \text{ is an } S\text{-semi-open-neighborhood of } A\}$.*
- (iv) *If $x \in X$, then $\{x\}$ is S -semi-closed.*
- (v) *If $x \neq y$, then there exists an S -semi-closed neighborhood of x to which y does not belong..*
- (vi) *If $x \in X$, then $\{x\} = \cap\{N : N \text{ is an } S\text{-semi-closed-neighborhood of } x\}$.*

Then

- (a) $S\text{-semi-}T_0 \iff (i) \iff (ii)$.
- (b) $S\text{-semi-}T_1 \iff (iii) \iff (iv)$.
- (c) $S\text{-semi-}T_2 \iff (v) \iff (vi)$.

Remark. Many properties of semi- T_i spaces (Maheshewari and Prasad 1975) can be easily obtained from the properties of S -semi- T_i spaces by setting $\tau^* = SO(X, \tau), i = 0, 1, 2$.

Theorem 4.2 *If X is an S -semi- T'_2 -space and $x \neq y$, then there exist two S -semi-open neighborhoods N_x of x and N_y of y such that $N'_x \cap N'_y = \Phi$ (N'_x is the set of all limit points of N_x).*

The converse of Theorem 4.2, is not true in general. But it is true in the case of $\tau^* = SO(X)$.

Theorem 4.3 *The property of being an S -semi- T_i -space is S -semi-topological property, $i = 0, 1, 2$.*

Definition 4.2 A function $f : X \rightarrow Y$ is said to be S -pre-semi-open if and only if the image of each S -semi-open set is an S -semi-open set.

Theorem 4.4 *The property of being S -semi- T_2 is preserved under S -pre-semi-open bijection.*

Remark. Since the intersection of an open set and an S -semi-open set is always S -semi-open, it follows that every open (S -semi-open) subspace of S -semi- T_0 (resp. S -semi- T_1 , S -semi- T_2 , S -semi- T_2') space is S -semi- T_0 (resp. S -semi- T_1 , S -semi- T_2 , S -semi- T_2') space.

5. S -SEMICLOSED GRAPHS AND STRONGLY S -SEMI-CLOSED GRAPHS

Definition 5.1 A subset A of the product space $X \times Y$ is S -semi-closed in $X \times Y$ if for each $(x, y) \in X \times Y - A$, there exist two S -semi-open neighborhoods U and V of x and y , respectively, such that $(U \times V) \cap A = \Phi$.

A function $f : X \rightarrow Y$ has an S -semi-closed graph, if the graph $G(f) = \{(x, f(x)) : x \in X\}$ is S -semi-closed in $X \times Y$.

Lemma 5.1 *A function $f : X \rightarrow Y$ has an S -semi-closed graph if and only if for each $x \in X, y \in Y$ such that $y \neq f(x)$, there exist two S -semi-open sets U and V containing x and y , respectively such that $f(U) \cap V = \Phi$.*

Proof. Necessity. Since f has an S -semi-closed graph, then for each $x \in X$ and $y \in Y$ such that $y \neq f(x)$, there exist two S -semi-open sets U and V containing x and y , respectively such that $(U \times V) \cap G(f) = \Phi$. This implies that for every $x \in U$ and $y \in V$, $f(x) \neq y$. So $f(U) \cap V = \Phi$.

Sufficiency. Let $(x, y) \notin G(f)$. Then there exist two S -semi-open sets U and V containing x and y , respectively such that $f(U) \cap V = \Phi$. This implies that for each $x \in U$ and $y \in V$, we have $f(x) \neq y$. So $(U \times V) \cap G(f) = \Phi$. Hence f has an S -semi-closed graph.

Theorem 5.1 *If $f : X \rightarrow Y$ is S -irresolute and Y is an S -semi- T_2 -space, then f has an S -semi-closed graph.*

Proof. Let $(x, y) \in (X \times Y) - G(f)$. Then $y \neq f(x)$, and since Y is S -semi- T_2 , there exist two S -semi-open sets U and V such that $f(x) \in U$, $y \in V$ and $U \cap V = \Phi$. Since f is S -irresolute, there exists an S -semi-open neighborhood W of x such that $f(W) \subseteq U$. Hence $f(W) \cap V = \Phi$. This implies that $(W \times V) \cap G(f) = \Phi$. Hence f has an S -semi-closed graph.

Theorem 5.2 *If $f : X \rightarrow Y$ is S -irresolute injection with an S -semi-closed graph, then X is S -semi- T_2 .*

Proof. Let $x_1, x_2 \in X$ such that $x_1 \neq x_2$. Then $f(x_1) \neq f(x_2)$. This implies that $(x_1, f(x_2)) \in (X \times Y) - G(f)$. Since f has an S -semi-closed graph, there exist two S -semi-open neighborhoods U and V of x_1 and $f(x_2)$, respectively such that $(U \times V) \cap G(f) = \Phi$. This gives $f(U) \cap V = \Phi$. Since f is S -irresolute, there exists an S -semi-open set W containing x_2 such that $f(W) \subseteq V$. Hence $f(W) \cap f(U) = \Phi$. Therefore $W \cap U = \Phi$ and X is an S -semi- T_2 -space.

Definition 5.2 A function $f : X \rightarrow Y$ has a strongly S -semi-closed graph if for each $(x, y) \notin G(f)$, there exist two S -semi-open sets U and V containing x and y , respectively, such that $(U \times sCl(V)) \cap G(f) = \Phi$.

Lemma 5.2 *A function $f : X \rightarrow Y$ has a strongly S -semi-closed graph if for each $(x, y) \notin G(f)$, there exist two S -semi-open neighborhoods U and V of x and y , respectively such that $f(U) \cap sCl(V) = \Phi$.*

Definition 5.3 Let (X, τ) be a topological space. Then a subset S of X is called an α -set if $S \subseteq Int[Cl(Int(S))]$.

Definition 5.4 A function $f : X \rightarrow Y$ has an α -closed graph if and only for each $x \in X, y \in Y$ such that $y \neq f(x)$, there exist two α -sets U and V in X and Y , respectively such that $x \in U$ and $y \in V$ such that $f(U) \cap Cl(V) = \Phi$.

Lemma 5.3 *A function $f : X \rightarrow Y$ has an α -closed graph if and only if for each $x \in X$ and $y \in Y$ such that $y \neq f(x)$, there exist an α -set U in X and an open set V in Y such that $x \in U$ and $y \in V$ satisfying $f(U) \cap Cl(V) = \Phi$.*

Proof. Necessity. Since f has an α -closed graph, then for each $x \in X$ and $y \in Y$ such that $y \neq f(x)$, there exist two α -sets U and V such that $x \in U, y \in V$ and $f(U) \cap Cl(V) = \Phi$. Let $W = Int[Cl(Int(V))]$. Then

$y \in W$ and W is open. Since V is an α -set, so clearly $Cl(W) \subseteq Cl(V)$. Thus $f(U) \cap Cl(W) = \Phi$.

Sufficiency. It follows from the fact that every open set is an α -set.

Corollary 5.1 *Suppose that $f : X \rightarrow Y$ has an α -closed graph. Then f has a strongly S -semi-closed graph.*

Proof. Let $x \in X$ and $y \in Y$ such that $f(x) \neq y$. Since f has an α -closed graph, then by Lemma 5.3, there exist an α -set U and an open set V such that $x \in U, y \in V$ and $f(U) \cap Cl(V) = \Phi$. Since every open set is S -semi-open, $sCl(V) \subseteq Cl(V)$, then $f(U) \cap sCl(V) = \Phi$. So f has a strongly S -semi-closed graph.

The connection between closed graphs, strongly closed graphs (Long and Herrington 1975), α -closed graphs and the new types of graphs can be shown from the following diagram:

$$\begin{array}{ccccc}
 \text{strongly closed graphs} & & \Rightarrow & & \text{closed graphs} \\
 \Downarrow & & & & \Downarrow \\
 \alpha\text{-closed graphs} & \Rightarrow & \text{strongly } S\text{-semi-closed graphs} & \Rightarrow & S\text{-semi-closed graphs}
 \end{array}$$

Theorem 5.3 *If $f : X \rightarrow Y$ is S -irresolute and Y is an S -semi- T'_2 -space, then f has a strongly S -semi-closed graph.*

Proof. Let $(x, y) \notin G(f)$. Then $y \neq f(x)$. Since Y is S -semi- T'_2 , there exist two S -semi-open neighborhoods U and V of y and $f(x)$, respectively, such that $Cl(U) \cap Cl(V) = \Phi$. Since f is S -irresolute, there exists an S -semi-open neighborhood W of x such that $f(W) \subseteq U \subseteq Cl(U)$. So $f(W) \cap Cl(V) = \Phi$. But $sCl(V) \subseteq Cl(V)$, then $f(W) \cap sCl(V) = \Phi$. Hence, f has a strongly S -semi-closed graph.

Theorem 5.4 *If $f : X \rightarrow Y$ is S -pre-semi-open and has an S -semi-closed graph, then f has a strongly S -semi-closed graph.*

Proof. Let $x \in X$ and $y \in Y$ such that $y \neq f(x)$. Since f has an S -semi-closed graph, there exist two S -semi-open neighborhoods U and V of x and y , respectively, such that $f(U) \cap V = \Phi$. Since f is S -pre-semi-open, then $f(U) \cap sCl(V) = \Phi$. Therefore f has a strongly S -semi-closed graph.

Theorem 5.5 *Let $f : X \rightarrow Y$ be a surjective function with a strongly S -semi-closed graph. Then Y is S -semi- T_2 -space.*

Proof. Let y_1 and y_2 be distinct points in Y . Then there exists an $x_1 \in X$ such that $f(x_1) = y_1$. Thus $(x_1, y_2) \notin G(f)$. Since f has a strongly S -semi-closed graph, then there exist two S -semi-open neighborhoods U and V of x_1 and y_2 respectively such that $f(U) \cap sCl(V) = \Phi$. Consequently $y_1 \notin sCl(V)$. So X is S -semi- T_2 -space.

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A MULTISTAGE TEST FOR DETECTING MULTIPLE OUTLIERS IN THE NORMAL CASE

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ABSTRACT. In this article, a multistage procedure for detecting multiple outliers is proposed. The proposed test statistic called $L_{n,k}$ used in this procedure is based on the generalized likelihood ratio principle. In each sequential step of the proposed procedure, $L_{n,k}$ is considered as a block test for testing multiple outliers in a random sample from a normal distribution with zero mean and unknown variance σ^2 . The test statistic $L_{n,k}$ is scale-invariant, and hence, it is suitable for any value of σ^2 . The test statistic $L_{n,k}$ is derived and compared with some well known single- and multiple outliers tests and found to be more powerful and less affected by the problems of swamping and masking. In addition, a real data example is worked out to demonstrate the use of the procedure.

1. INTRODUCTION

The problem of detecting outliers is old, yet it is of considerable importance in applied statistics. Many authors use “outlier” to indicate any observation that does not come from the target distribution. The literature on outliers is vast and the books by Barnett and Lewis (1978) and Hawkins (1980) and the paper by Beckman and Cook (1983) provide useful surveys of much of the relevant literature.

Outlier detection problems associated with the normal distributions have been generally treated as problems in hypotheses testing. The null hypothesis is generally that the set of all observations is a random sample from some

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normal distribution with unknown mean and unknown variance, while the alternative hypothesis is that some of the observations came from different (in mean or variance or both) normal distributions. Most outlier detection procedures depend on location-scale invariant test statistics to accommodate the general normality case. Many formal techniques for the identification of outliers have been proposed in the literature, some of which are single-outlier procedures that are designed to test for only one outlier. Other methods are multiple outliers procedures that can detect more than one outlier in the sample. Multiple outliers procedures generally include the sequential application of a single outlier test to test for more than one outlier. Barnett and Lewis (1978) call the repeated application of single outlier tests for rejection of multiple outliers a consecutive test. Another type of multiple outliers tests, called block tests by Barnett and Lewis (1978), are used to test sets of data simultaneously in one step. The masking effect and the swamping effect are two problems associated with multiple outliers tests. The masking effect is the inability of a multiple outlier's procedure to identify even a single outlier in the presence of several potential outliers. This problem usually happens when outliers of similar magnitude form subgroups in the sample. This phenomenon makes procedures based on the sequential application of one-outlier tests insensitive. This problem also occurs with block tests when the true number of outliers is underestimated. On the other hand, the swamping effect is the tendency of a multiple outlier's procedure to declare more outliers than there are in the data. Many procedures for detecting multiple outliers suffer from the swamping effect when the maximum number of potential outliers assumed is much larger than the true number of outliers.

We propose a new statistic called $L_{n,k}$ for detecting multiple outliers. In each step of the proposed procedure, $L_{n,k}$ is considered as a block test for testing multiple outliers in a random sample from a normal distribution with zero mean and unknown variance σ^2 . The test statistic $L_{n,k}$ is scale-invariant, and hence, the procedure is suitable for any value of σ^2 .

In Section 2, we present a brief review for some early multiple outliers tests. In Section 3, we derive the test statistic $L_{n,k}$, describe the proposed multistage procedure for detecting multiple outliers, and compare it with some well known single-and multiple outliers tests. In Section 4, we present an example to

demonstrate the use of the proposed multistage procedure.

2. REVIEW OF SOME EARLY WORK ON OUTLIERS DETECTION

Several procedures for detecting outlying observations in the normal case can be found in literature. Since we are concerned about procedures detecting multiple outliers, our discussion will be mainly about these types of procedures.

Dixon (1950) proposed criteria for detecting one outlier. His criteria are ratios of the distance between a suspected observation and its nearest neighbor to the range of the sample (with zero, one, or two observations omitted). Dixon (1953) has suggested a sequential application of his criteria for detecting multiple outliers.

Grubbs (1950) proposed some statistics to test the significance of the largest observation, the smallest observation, the two largest observations, the two smallest observations, and the largest and smallest observations. In the case of one outlier, Grubbs's statistic is statistically equivalent to the test statistic called Extreme Studentized Deviate (ESD) proposed by Pearson and Chandra Sekar (1936). Grubbs derived for the first time the exact distribution of ESD. ESD has been shown that it maximizes the probability of making the correct decision under the mean-shift model in the case of one outlier by Paulson (1952), and Kudo (1956), and hence, it is optimal in this sense. Ferguson (1961) showed the same property of ESD when one outlier is generated under the variance-inflation model. Therefore, we shall choose the ESD test in our comparison study that will be presented in this paper.

Ferguson (1961) has shown that tests based on the sample kurtosis are locally best invariant test for testing small shifts in the mean in either direction. He also showed that tests based on the sample kurtosis are locally best invariant test for testing small positive shifts in the variance. Ferguson (1961) suggested the sequential application of the kurtosis test to test for multiple outliers. Since tests based on the sample kurtosis were shown to be optimal, we shall include the test based on the sequential application of the kurtosis in our comparison study.

Tietjen and Moore (1972) generalized Grubbs's statistics to a class of statistics that can detect exactly k outliers. They proposed two Grubbs-type statistics for the detection of multiple outliers. For testing the k largest observations of the sample $x_1, x_2, \dots, x_n$, they proposed the test statistic $L_k = \sum_{i=1}^{n-k} (x_{(i)} - \bar{x}_k)^2 / \sum_{i=1}^n (x_i - \bar{x})^2$ where $x_{(i)}$ is the i -th smallest order statistic, $\bar{x}_k = \sum_{i=1}^{n-k} x_{(i)} / (n-k)$, and $\bar{x}$ is the mean of the full sample. For testing the k smallest observations they proposed the test statistic $L_k^* = \sum_{i=k+1}^n (x_{(i)} - \bar{x}_k^*)^2 / \sum_{i=1}^n (x_i - \bar{x})^2$ where $\bar{x}_k^* = \sum_{i=k+1}^n x_{(i)} / (n-k)$. For testing exactly k outliers in the sample, they proposed the test statistic $E_k = \sum_{i=1}^{n-k} (z_{(i)} - \bar{z}_k)^2 / \sum_{i=1}^n (z_i - \bar{z})^2$ where $\bar{z}_k = \sum_{i=1}^{n-k} z_{(i)} / (n-k)$, and z_i is the observation whose absolute residual, $|z_i - \bar{z}|$, is the i -th largest. Tietjen and Moore suggested estimating k by using the largest gap in the order sample when using L_k and L_k^* . They did not suggest any method to estimate k when using E_k . Hawkins (1979) showed that even if k is correctly identified, E_k might detect wrong observations as outliers in the case where the outliers occur on one side of the data. For this reason, we choose not to include it in our comparison study.

Rosner (1975) developed multiple outliers test statistics which require only the knowledge of the maximum number of possible outliers, k . Hawkins (1978) and Cook and Beckman (1980) showed that Rosner's procedure suffers from the swamping effect, i.e., it tends to declare more outliers than there are in the sample. This particular problem is also confirmed by our simulation study when we compared Rosner's procedure with the $L_{n,k}$ procedure (not shown here). For this reason, we choose not to include it in our comparison study.

The proposed $L_{n,k}$ procedure will be compared with the sequential procedures based on ESD and KUR test statistics for detecting nonzero means in a random sample from a normal distribution with unknown variance.

3. DEVELOPMENT OF THE $L_{n,k}$ PROCEDURE

Suppose $x_1, x_2, \dots, x_n$ are n observations which, under a null hypothesis, constitute a random sample from $N(0, \sigma^2)$. The alternative hypothesis is that m of these observations have nonzero means and they are considered as outliers. That is, $x_1, x_2, \dots, x_n$ is a sample composed of two subsets

$$J_1 = \{x_{i_1}, x_{i_2}, \dots, x_{i_{n-m}}\} \text{ and } J_2 = \{x_{i_{n-m+1}}, x_{i_{n-m+2}}, \dots, x_{i_n}\},$$

where $x_{i_j} \sim N(0, \sigma^2)$, $j = 1, 2, \dots, n - m$, $x_{i_j} \sim N(\theta_{i_j}, \sigma^2)$, $j = n - m + 1, n - m + 2, \dots, n$, and m is some integer such that $0 \leq m < n$. The subset J_1 is called the set of "inliers" and the subset J_2 is called the set of "outliers". Our purpose is to identify the subsets J_1 and J_2 .

Derivation of the Test Statistic $L_{n,k}$

The inference problem of detecting multiple outliers is formalized as follows. Suppose $x_1, x_2, \dots, x_n$ are independent random variables such that $x_i \sim N(\theta_i, \sigma^2)$, $i = 1, 2, \dots, n$, where all parameters are unknown. Let k be the maximum number of possible outliers such that $0 < k < n$. Let $I = \{1, 2, \dots, \binom{n}{k}\}$, where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. Let $F_k = \{A_{i(k)} : i \in I\}$ be the collection of all different subsets of size k of the distinct elements of $\{1, 2, \dots, n\}$. Testing that the x_j 's with $j \in A_{i(k)}$ for some $i \in I$ are outliers in the sample $x_1, x_2, \dots, x_n$ is equivalent to testing:

$$(3.1) \quad \begin{aligned} H_0 : & \quad \theta_1 = \theta_2 = \dots = \theta_n = 0, \text{ against} \\ H_1^{(i)} : & \quad \theta_j \neq 0 \text{ for } j \in A_{i(k)}, \text{ and } \theta_j = 0 \text{ for } j \notin A_{i(k)}. \end{aligned}$$

The generalized likelihood ratio test (G.L.R.T.) statistic for testing (3.1) is:

$$\lambda_i(x, k) = \frac{\sup_{H_0} L(\theta_1, \theta_2, \dots, \theta_n, \sigma^2 | x)}{\sup_{H_1^{(i)}} L(\theta_1, \theta_2, \dots, \theta_n, \sigma^2 | x)},$$

where

$$L(\theta_1, \theta_2, \dots, \theta_n, \sigma^2 | x) = (2\pi\sigma^2)^{-n/2} \exp \left\{ - \sum_{i=1}^n (x_i - \theta_i)^2 / (2\sigma^2) \right\}.$$

It can be easily shown that

$$\lambda_i(x, k) = \left(1 + \frac{\sum_{j \in A_i(k)} x_j^2}{\sum_{j \notin A_i(k)} x_j^2} \right)^{-n/2}$$

The null hypothesis in (3.1) is rejected for small values of $\lambda_i(x, k)$ or equivalently it is rejected for large values of the following statistic:

$$T_{i(k)} = \frac{\sum_{j \in A_i(k)} x_j^2/k}{\sum_{j \notin A_i(k)} x_j^2/(n-k)}.$$

Let $S_{i(k)} = \sum_{j \in A_i(k)} x_j^2/\sigma^2$ and $U_{i(k)} = \sum_{j \notin A_i(k)} x_j^2/\sigma^2$. Under H_0 , $S_{i(k)}$ and $U_{i(k)}$ are independent Chi-square random variables with k and $n - k$ degrees of freedom, respectively. Therefore, $T_{i(k)} = \frac{S_{i(k)}/k}{U_{i(k)}/(n-k)}$ has F -distribution with k and $n - k$ degrees of freedom for all $i \in I$.

Now, to test for presence of exactly k outliers in the sample $x_1, x_2, \dots, x_n$, we test:

$$(3.2) \quad \begin{aligned} H_0 : & \theta_1 = \theta_2 = \dots = \theta_n = 0, \text{ against} \\ H_1 : & \text{ exactly } k \text{ of the } \theta_i \text{'s are not equal to } 0. \end{aligned}$$

The G.L.R.T. statistic for testing (3.2) is given by:

$$\begin{aligned} \lambda(x, k) &= \frac{\sup_{H_0} L(\theta_1, \theta_2, \dots, \theta_n, \sigma^2 | x)}{\max_{i \in I} \sup_{H_1^{(i)}} L(\theta_1, \theta_2, \dots, \theta_n, \sigma^2 | x)} \\ &= \min_{i \in I} \lambda_i(x, k) \\ &= \left(1 + \frac{k}{n-k} \max_{i \in I} T_{i(k)} \right)^{-n/2}. \end{aligned}$$

H_0 is rejected for small values $\lambda(x, k)$, or equivalently it is rejected for large values of the following statistic:

$$(3.3) \quad L_{n,k} = \max_{i \in I} T_{i(k)}.$$

It is difficult to determine the null distribution of $L_{n,k}$ analytically even though $T_{i(k)} \sim F_{(k,n-k)}$ for all $i \in I$ because the $T_{i(k)}$'s are not independent random variables. Therefore, the percentiles of the null distribution of $L_{n,k}$ were empirically simulated. Table 3.1 gives some simulated percentiles of $L_{n,k}$ for some values of n and k .

Table 3.1: Some simulated percentiles of $L_{n,k}$

(1- α) percentile of $L_{n,k}$					(1- α) percentile of $L_{n,k}$				
		α					α		
n	k	0.01	0.05	0.10	n	k	0.01	0.05	0.10
25	5	16.067	12.101	10.487	24	5	16.438	12.439	10.698
	4	15.233	11.516	10.037		4	15.437	11.719	10.161
	3	14.868	11.172	9.772		3	14.930	11.289	9.818
	2	15.132	11.159	9.709		2	15.128	11.327	9.727
	1	16.860	11.994	10.173		1	17.298	11.988	10.159
23	5	17.202	12.759	10.880	22	5	17.959	13.092	11.159
	4	15.983	11.975	10.309		4	16.591	12.175	10.463
	3	15.259	11.412	9.954		3	15.799	11.537	9.997
	2	15.277	11.369	9.778		2	15.881	11.564	9.748
	1	17.120	12.034	10.112		1	17.044	12.024	10.020

Let $y_i = x_i^2 (i = 1, 2, \dots, n)$ and $y_{(i)}$ be the i -th smallest order statistic of the random sample $y_1, y_2, \dots, y_n$. It can be shown that the statistic $L_{n,k}$ can be written as:

$$(3.4) \quad L_{n,k} = \frac{\sum_{j=n-k+1}^n y_{(j)}/k}{\sum_{j=1}^{n-k} y_{(j)}/(n-k)}.$$

The formula in (3.4) is easier than the formula in (3.3) for calculating $L_{n,k}$ for a given sample. Since in the successive stages of the proposed procedure, the test statistic $L_{n,k}$ is calculated with values of n and k reduced by one at each stage, the following recursive formula is useful in applying this procedure:

$$L_{n-j,k-j} = \frac{1}{k-j} \left[(k-j+1)L_{n-j+1,k-j+1} - \frac{y_{(n-j+1)}}{\bar{D}_{n-k}} \right]; j = 1, 2, \dots, k-1,$$

where $\bar{D}_{n-k} = \sum_{i=1}^{n-k} y_{(i)}/(n-k)$.

3.2 Selection of k

The true number of the outliers, r , is generally unknown and cannot be specified in advance. Therefore, to apply an appropriate outlier detection procedure, a prior estimate, k , of r is required. If k is selected in advance and is smaller than the true number of outliers present, masking effect may cause the risk of not detecting any outliers even if they are present. On the other hand, if k is too large, swamping effect may lead to declaring valid observations as outliers. In our procedure, the initial estimate for r is denoted by k , which will be carefully chosen, is the number used to calculate $L_{n,k}$.

Marasinghe (1985), in identification of outliers in regression problem, points out that tests based on block statistics have enough power to reject the no-outlier hypothesis even when k is overestimated. Also, our simulation results show that the reduction in the power of the multiple outliers test using $L_{n,k}$ statistic is small when k is chosen to be slightly larger than the actual number of outliers, r (see Table 3.4). To estimate k , we propose to use the half-normal plot. In the half-normal plot the “inlier” observations tend to fall on a straight line passing through the origin and with a slope approximately equal to $1/\sigma$, while the outlying observations tend to fall “far” away from this line. Usually, a number k_0 ($k_0 \geq 0$) of the observations clearly appear far away from the line, while a number k_1 ($k_1 \geq 0$) of the observations may or may not fall on that line. We suggest that one should overestimate r by $k = k_0 + k_1$. This procedure means that any suspected observation will also be treated as an outlier in the initial process of estimating r by k .

3.3 The Multistage Procedure for Detecting Multiple Outliers

The proposed multistage procedure takes advantage of the property that tests based on block statistics like $L_{n,k}$ have enough power to reject the no-outlier hypothesis even when k is overestimated. It is based on using $L_{n,k}$ repeatedly. At each stage, one observation from the sample is deleted. The proposed procedure allows the percentage points of $L_{n,k}$ to be used repeatedly.. Marasinghe (1985) has used a similar approach for detecting multiple outliers in linear regression. Marasinghe (1985) points out that John and Draper (1978) have used a similar approach to his for detecting multiple outliers in two-way tables.

The nested sequence of hypotheses tested by the proposed multistage procedure can be reformulated as testing the following sequence of hypotheses. First, the null $H_0^{(1)}$: no-outliers versus $H_1^{(1)}$: at most k outliers is tested using $L_{n,k}$. If $H_0^{(1)}$ is rejected, the observation that corresponds to the largest squared value is removed and $H_0^{(2)}$: no-outliers versus $H_1^{(2)}$: at most $(k-1)$ outliers is tested using $L_{n-1,k-1}$, computed from the remaining sample. This process is continued until a null hypothesis $H_0^{(m+1)}$: no-outliers versus $H_1^{(m+1)}$: at most $(k-m)$ outliers fails to be rejected ($m \leq k$). This procedure assumes that the observation removed in each stage is a true outlier, in which case the Type I error rate at each stage is the level specified for the test. In this procedure, the percentage points of $L_{n,k}$ can be used in successive stages with values of n and k reduced by one at each stage. The final estimate of r obtained by this sequential procedure is actually m and the declared outliers are those observations that correspond to $y_{(n)}, y_{(n-1)}, \dots, y_{(n-m+1)}$. In applying this procedure, and in order not to miss potential outliers, we suggest using a large significance level such as $\alpha = 0.1$.

3.4 Power Comparison Between $L_{n,1}$, ESD, and KUR for Detecting One Outlier

We have mentioned earlier that the procedures of detecting one outlier based on the statistics ESD and KUR have some optimal properties. In this section, we empirically compare powers of these tests to the power of $L_{n,1}$ for some chosen sample sizes. The power of a procedure for detecting outliers is defined as the ratio of the number of the samples detecting the correct number of outliers to the total number of samples. The statistics ESD, and KUR are defined as follows:

$$ESD = \frac{\max_{i=1, \dots, n} |x_i - \bar{x}|}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 / n}},$$

$$KUR = \frac{n \sum_{i=1}^n (x_i - \bar{x})^4}{\left[\sum_{i=1}^n (x_i - \bar{x})^2 \right]^2},$$

where $\bar{x} = \sum_{i=1}^n x_i/n$.

First, we compare the empirical powers for detecting one outlier in the presence of exactly one outlier in the sample. This comparison was performed for $n = 10$, and 20 , and for $\alpha = 0.01$, and 0.05 . In this simulation study we consider $x_i \sim N(0, 1), i = 1, 2, \dots, n-1$, and $x_n \sim N(\lambda, 1)$, and $x_1, x_2, \dots, x_n$ are independent, and $\lambda = 0, 1, \dots, 10$. For each combination of n, α , and λ , we independently simulated 10,000 random samples from $N(0, 1)$. For each random sample, the outlier was introduced by adding known quantity λ to the last observation and computed the values of $L_{n,1}$ ESD, and KUR. The empirical power of each test is then the total number of the random samples that rejected the no-outlier hypothesis divided by 10,000. The results of this study are presented in Table 3.2 and Figure 3.1. This simulation study shows that $L_{n,1}$ is more powerful than both ESD and KUR for almost all combinations of n, α , and λ .

Second in order to investigate the masking effect on each of the tests $L_{n,1}$, ESD, and KUR, we compare the empirical powers for detecting one outlier in presence of exactly two outlier in the sample. This comparison was performed for $n = 20$ and $\alpha = 0.05$. In this simulation study we consider $x_i \sim N(0, 1), i = 1, 2, \dots, n-2, x_{n-1} \sim N(\lambda_1, 1)$, and $x_n \sim N(\lambda_2, 1)$, and $x_1, x_2, \dots, x_n$ are independent. We chose $\lambda_2 = 5, 6, 7$, and 8 , and for each value of λ_2 , we used $\lambda_1 = 0, 2, 3, \dots, \lambda_2$. For each combination of n, λ_1 , and λ_2 , we independently simulated 10,000 random samples from $N(0, 1)$. For each random sample, the outliers were introduced by adding known quantity λ_1 to the $(n-1)$ -th observation and by adding known quantity λ_2 to the n -th observation and computed the values of $L_{n,1}$, ESD, and KUR. The empirical power of each test is then the total number of the random samples that rejected the no-outlier hypothesis divided by 10,000. The results of this study are presented in Table 3.3 and Figure 3.2. This simulation study shows that $L_{n,1}$ is less effected by the masking problem and more powerful than ESD for almost all combinations of n, λ_1 , and λ_2 . It is also true that $L_{n,1}$ is less effected by the masking problem and slightly more powerful than KUR for most of the combinations of n, λ_1 , and λ_2 .

Table 3.2: Empirical Powers of $L_{n,1}$, ESD , and KUR for Detecting One Outlier in Presence of One Outlier (This Table Is Illustrated By Figure 3.1)

		$\alpha = 0.01$			$\alpha = 0.05$		
n	λ	$L_{n,1}$	ESD	KUR	$L_{n,1}$	ESD	KUR
10	0.0	0.0118	0.0117	0.0111	0.0534	0.0526	0.0536
	1.0	0.0131	0.0134	0.0106	0.0613	0.0533	0.0533
	2.0	0.0406	0.0302	0.0277	0.1300	0.1031	0.1028
	3.0	0.1515	0.1062	0.0955	0.3433	0.2679	0.2616
	3.5	0.2483	0.1779	0.1649	0.4865	0.3867	0.3787
	4.0	0.3705	0.2739	0.2581	0.6463	0.5300	0.5187
	4.5	0.4925	0.3698	0.3512	0.7713	0.6655	0.6501
	5.0	0.6251	0.4997	0.4761	0.8633	0.7716	0.7544
	5.5	0.7403	0.6051	0.5810	0.9286	0.8584	0.8463
	6.0	0.8365	0.7153	0.6887	0.9686	0.9245	0.9157
	6.5	0.9092	0.8106	0.7890	0.9866	0.9589	0.9536
	7.0	0.9496	0.8786	0.8632	0.9956	0.9829	0.9789
	8.0	0.9886	0.9568	0.9472	0.9993	0.9972	0.9964
	9.0	0.9985	0.9889	0.9864	1.0000	0.9999	0.9998
	10.0	0.9998	0.9979	0.9969	1.0000	0.9999	0.9998
20	0.0	0.0112	0.0105	0.0080	0.0505	0.0508	0.0477
	1.0	0.0141	0.0136	0.0119	0.0567	0.0540	0.0523
	2.0	0.0448	0.0418	0.0375	0.1263	0.1173	0.1144
	3.0	0.1715	0.1528	0.1370	0.3632	0.3256	0.3166
	3.5	0.3105	0.2790	0.2497	0.5383	0.4914	0.4764
	4.0	0.4713	0.4307	0.3977	0.7073	0.6641	0.6429
	4.5	0.6322	0.5837	0.5446	0.8333	0.7929	0.7752
	5.0	0.7704	0.7248	0.6942	0.9153	0.8846	0.8707
	5.5	0.8730	0.8364	0.8105	0.9647	0.9524	0.9430
	6.0	0.9428	0.9224	0.9045	0.9874	0.9809	0.9756
	6.5	0.9756	0.9630	0.9519	0.9956	0.9927	0.9917
	7.0	0.9915	0.9851	0.9802	0.9994	0.9986	0.9975
	8.0	0.9993	0.9985	0.9975	1.0000	1.0000	1.0000
	9.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	10.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Figure 3.1: Empirical Power of $L_{n,1}$, ESD , and KUR for Detecting One Outlier in Presence of One Outlier ($n = 10, 20$, and $\alpha = 0.01, 0.05$)

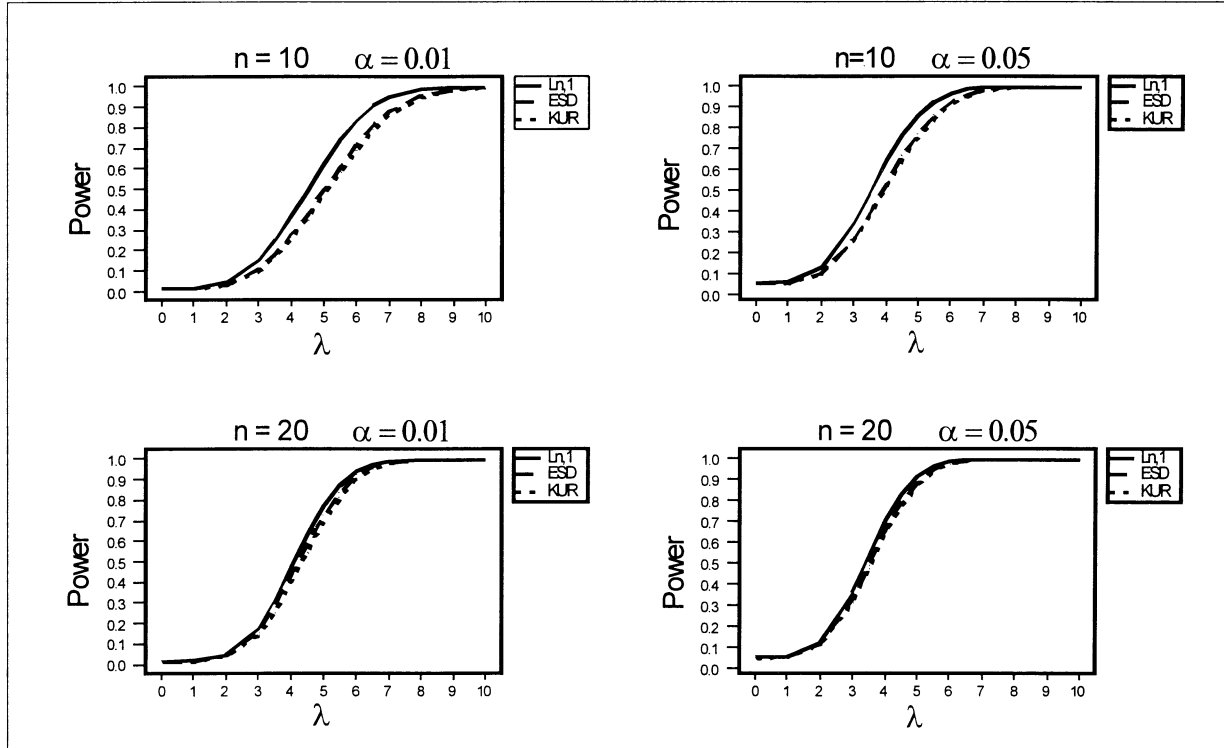
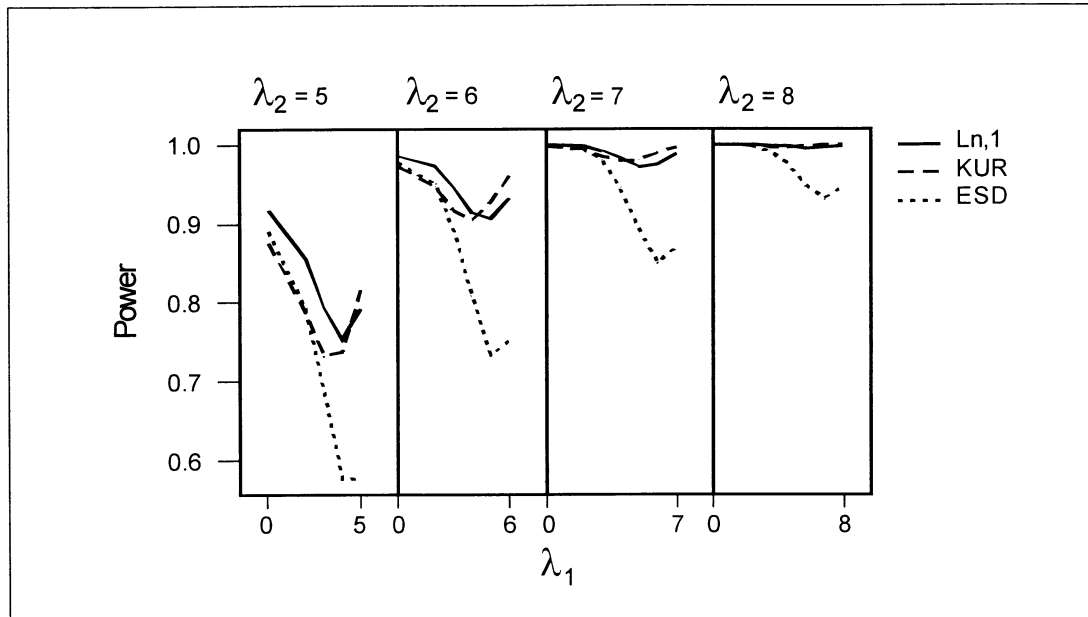


Table 3.3: Empirical Powers of $L_{n,1}$, ESD , and KUR for Detecting One Outlier in Presence of Two Outlier for The Case where $n = 20$ and $\alpha = 0.05$ (This Table Is Illustrated By Figure 3.2)

λ_1	λ_2	$L_{n,1}$	ESD	KUR
0	5	0.9165	0.8902	0.8752
2		0.8544	0.7934	0.7867
3		0.7933	0.6874	0.7327
4		0.7534	0.5764	0.7378
5		0.7911	0.5781	0.8153
0	6	0.9851	0.9771	0.9731
2		0.9726	0.9497	0.9460
3		0.9447	0.8923	0.9161
4		0.9134	0.8088	0.9046
5		0.9058	0.7330	0.9264
6		0.9313	0.7510	0.9603
0	7	0.9990	0.9986	0.9981
2		0.9976	0.9946	0.9935
3		0.9916	0.9799	0.9844
4		0.9824	0.9446	0.9790
5		0.9717	0.8921	0.9803
6		0.9741	0.8495	0.9885
7		0.9876	0.8694	0.9960
0	8	0.9999	0.9999	0.9999
2		0.9999	0.9997	0.9997
3		0.9996	0.9976	0.9984
4		0.9980	0.9911	0.9966
5		0.9971	0.9754	0.9967
6		0.9936	0.9474	0.9974
7		0.9960	0.9311	0.9991
8		0.9981	0.9459	0.9992

Figure 3.2: Empirical Power of $L_{n,1}$, ESD , and KUR for Detecting One Outlier in Presence of Two Outlier ($n = 20$, and $\alpha = 0.05$)



3.5 Power Comparisons Between $L_{n,k}$ and Sequential Applications of $L_{n,1}$, ESD, and KUR for Detecting Multiple Outliers

In this section, we compare the empirical performance of the new multistage multiple outliers procedure with the sequential application of the tests based on the statistics $L_{n,1}$, ESD, and KUR. The processes of generating the random samples and introducing the outliers used in this section are similar to those used in Section 3.4. In this simulation study, we used $n = 25, \alpha = 0.05$. The performance of each procedure when at most 3 outliers were present in the sample was evaluated. The empirical power of a test is the number of random samples in which the test detects the correct number of outliers divided by the total numbers of samples. The results of this study are presented in Table 3.4.

The numbers presented in Table 3.4 are the percentages of trials in which the indicated number of outliers (m) was detected by each procedure. The patterns of contamination including a no-outlier case used in this study are also shown in Table 3.4. The effect of using different initial values of k in the multistage procedure was examined by repeating the procedure for $k = 1, 2, 3, 4$, and 5.

From this simulation study we conclude several results as follows. First, the empirical size of the test of the new procedure is almost equal to the specified significance level, viz. $\alpha = 0.05$, whatever initial value is chosen for k . This indicates that the new procedure adequately controls Type I error. Second, for small λ_1 (i.e., for the case (5,0,0)), the power of the new procedure, when only one outlier is present, is slightly less than that of sequential $L_{n,1}$, ESD, and KUR only when the true number of outliers are overestimated, (i.e., when using $k \geq 2$) and it decreases slightly for large values of k . For large values of λ_1 (i.e., for the case (10,0,0)), the power of the new procedure is almost greater or equal to those of sequential $L_{n,1}$, ESD, and KUR for all values of k . Third, the power of detecting multiple outliers of the $L_{n,k}$ procedure is always larger than those of ESD, and KUR when the initial values of k , are chosen equal to or larger than the true number of outliers. Fourth, the power of detecting multiple outliers based on the sequential $L_{n,1}$ procedure is always larger than those of ESD, and KUR. Fifth, when r is correctly estimated or slightly overestimated, the $L_{n,k}$ multistage procedure is always more powerful than the sequential $L_{n,1}$

procedure, while the sequential $L_{n,1}$ is slightly more powerful only when r is much overestimated. Sixth, when k is correctly chosen, the new procedure is more powerful than the sequential $L_{n,1}$, ESD, and KUR for all cases. Seventh, the new procedure is not effected by the problem of swamping, whereas the sequential procedures based on ESD and KUR highly suffer from this problem, see for example the cases $(10,5,0)$, $(10,10,0)$ and $(15,10,5)$. Eighth, in detecting multiple outliers and when r is correctly estimated or slightly overestimated, the $L_{n,k}$ multistage procedure is generally less effected by the problem of masking than the sequential procedures based on $L_{n,1}$, ESD and KUR, see for example the cases $(5,5,5)$ and $(5,5,-5)$. Finally, this result shows that the power of detecting multiple outliers using the new multistage procedure remains high when the initial value of k is chosen equal to or larger than r . Moreover, the decrease in the power when the initial value of k is chosen slightly larger than r is small.

Table 3.4: Performance of Multiple outliers Procedures ($n = 25, \alpha = 0.05$)

No. of Outliers (r)	Outlier Pattern ($\lambda_1, \lambda_2, \lambda_3$)	No. of Outliers Found (m)	Sequential procedures			Multistage procedure Using $L_{n,k}$				
			ESD	KUR	$L_{n,1}$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
0	(0,0,0)	0	95.0	94.8	95.0	95.0	95.1	95.2	95.1	95.0
		≥ 1	5.0	5.2	5.0	5.0	4.9	4.8	4.9	5.0
1	(5,0,0)	0	9.8	10.8	8.0	8.0	11.2	17.0	22.1	26.6
		1	85.5	84.0	87.3	92.0	83.8	78.0	72.6	68.1
		≥ 2	4.7	5.2	4.7	—	5.0	5.0	5.3	5.3
1	(10,0,0)	0	0	0	0	0	0	0	0	0
		1	95.2	95.2	95.2	100	95.2	95.6	95.2	95.1
		≥ 2	4.8	4.8	4.8	—	4.8	4.4	4.8	4.9
2	(5,5,0)	0	23.1	8.5	11.3	11.3	1.9	2.8	4.1	6.1
		1	11.3	13.2	11.6	88.7	13.7	19.2	23.4	28.5
		2	40.6	48.8	73.0	—	84.4	73.1	67.6	60.6
		≥ 3	25.0	29.5	4.1	—	—	4.9	4.9	4.8
2	(10,5,0)	0	0	0	0	0	0	0	0	0
		1	10.1	11.4	8.5	100	8.5	12.5	16.9	22.8
		2	51.7	51.5	86.8	—	91.5	82.6	78.3	72.7
		≥ 3	38.2	37.1	4.7	—	—	4.9	4.8	4.5
2	(10,10,0)	0	0	0	0	0	0	0	0	0
		1	0	0	0	100	0	0	0	0
		2	60.5	60.6	94.9	—	100	94.9	95.1	95.5
		≥ 3	39.5	39.4	5.1	—	—	5.1	4.9	4.5
2	(5,-5,0)	0	9.1	2.2	11.0	11.0	1.8	2.5	3.6	5.2
		1	14.4	17.9	11.5	89.0	13.6	19.0	24.4	29.2
		2	47.7	49.7	73.6	—	84.6	73.6	66.8	61.0
		≥ 3	28.8	30.2	3.9	—	—	4.9	5.2	4.6
2	(10,-10,0)	0	0	0	0	0	0	0	0	0
		1	0	0	0	100	0	0	0	0
		2	59.7	59.6	94.3	—	100	94.3	94.7	95.0
		≥ 3	40.3	40.4	5.7	—	—	5.7	5.3	5.0
3	(5,5,5)	0	69.0	25.7	46.2	46.2	4.2	1.1	1.1	1.6
		1	10.1	5.8	10.3	53.8	21.1	4.1	4.6	6.6
		2	3.9	8.5	7.2	—	74.7	15.4	21.5	29.7
		3	7.3	31.8	34.4	—	—	79.4	67.8	56.7
		≥ 4	9.7	28.2	1.9	—	—	—	5.0	5.4
3	(10,5,5)	0	0.2	0	0	0	0	0	0	0
		1	25.4	9.5	13.6	100	13.6	2.1	2.6	4.0
		2	9.6	10.4	10.4	—	86.4	12.6	18.1	25.9
		3	31.0	38.2	72.3	—	—	85.3	74.6	65.1
		≥ 4	33.8	41.9	3.7	—	—	—	4.7	5.0
3	(10,10,5)	0	0	0	0	0	0	0	0	0
		1	0	0	0	100	0	0	0	0
		2	6.9	7.4	8.3	—	100	8.3	12.3	18.7
		3	3.6	4.0	87.2	—	—	91.7	83.0	76.1
		≥ 4	89.5	88.6	4.5	—	—	—	4.7	5.2
3	(5,5,-5)	0	44.9	4.0	46.4	46.4	4.7	1.0	1.2	1.8
		1	13.2	8.5	10.3	53.6	21.2	4.4	5.0	6.9
		2	7.0	12.7	7.5	—	74.1	15.6	22.2	29.7
		3	15.8	35.8	34.0	—	—	79.0	66.6	56.3
		≥ 4	19.1	39.0	1.8	—	—	—	5.0	5.3
3	(5,5,-10)	0	0	0	0	0	0	0	0	0
		1	23.8	9.6	12.8	100	12.8	2.1	2.7	4.1
		2	9.9	10.5	10.7	—	87.2	13.0	18.3	25.6
		3	32.5	38.4	72.4	—	—	84.9	73.9	65.0
		≥ 4	33.7	41.6	4.0	—	—	—	5.0	5.3
3	(10,5,-10)	0	0	0	0	0	0	0	0	0
		1	0	0	0	100	0	0	0	0
		2	8.3	9.2	8.0	—	100	8.0	11.9	18.1
		3	18.4	18.4	87.5	—	—	92.0	83.3	76.8
		≥ 4	73.3	72.4	4.5	—	—	—	4.8	5.1
3	(15,15,-15)	0	7.4	0	9.4	9.4	0	0	0	0
		1	0	0	0	90.6	0	0	0	0
		2	0	0	0	—	100	0	0	0
		3	50.2	55.7	86.2	—	—	100	95.1	94.9
		≥ 4	42.4	44.3	4.4	—	—	—	4.9	5.1

4. AN EXAMPLE

To demonstrate the use of the multistage procedure proposed for detecting multiple outliers, we present the following data. Daniel (1959) reported the results of a 2^5 factorial experiment where the 31 contrasts arranged in order of increasing absolute value as follows.

0.0000	0.0281	-0.0561	-0.0842	-0.0982	0.1263	0.1684	0.1964
0.2245	-0.2526	0.2947	-0.3087	0.3929	0.4069	0.4209	0.4350
0.4630	-0.4771	0.5472	0.6595	0.7437	-0.7437	-0.7577	-0.8138
-0.8138	-0.8980	1.0800	-1.3050	2.1470	-2.6660	-3.1430	

The half-normal probability plot of this data is shown in Figure 4.1. Table 4.1 summarizes the calculations for four possible selections of k_0 and k_1 based on the half-normal probability plot. We realize that the results of the test coincide for all cases in declaring three outliers at $\alpha = 0.10$. The three outliers declared are 2.147, -2.666, and -3.143. This result supports our claim that slightly overestimating the true number of outliers, r , by the initial estimate k will not cause a much decrease in the power of the test.

Figure 4.1: The half-normal probability plot

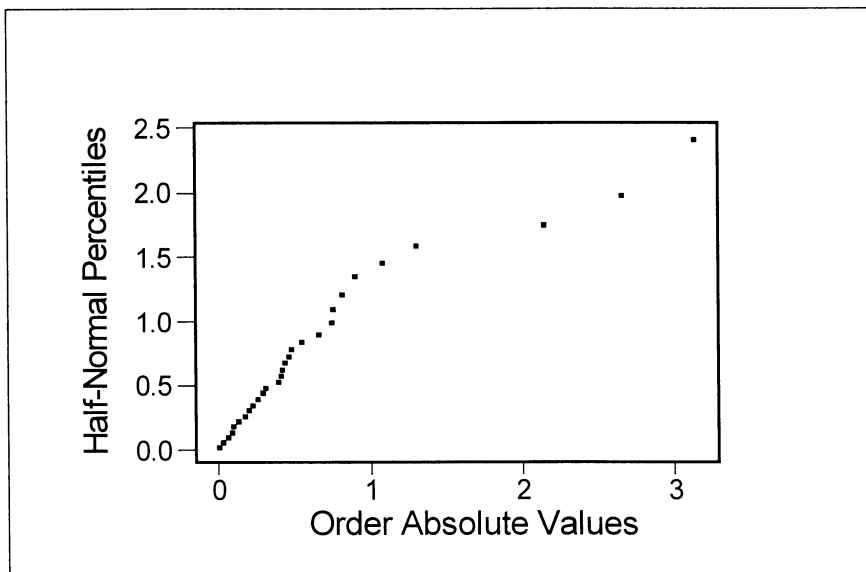


Table 4.1: Calculation of the test

				Simulated Critical Values of $L_{n,k}$		
	n	k	$L_{n,k}$	$\alpha=0.05$	$\alpha=0.10$	
a. $k_0=3$	31	3	22.5376**	10.760	9.467	
	$k_1=0$	30	2	18.3424**	11.043	9.625
	$k=3$	29	1	14.4320**	12.026	10.151
b. $k_0=3$	31	4	21.7211**	10.848	9.572	
	$k_1=1$	30	3	16.6821**	10.754	9.491
	$k=4$	29	2	11.7704**	11.075	9.699
		28	1	6.3509	11.999	10.131
c. $k_0=3$	31	5	20.9453**	11.103	9.830	
	$k_1=2$	30	4	15.6101**	10.879	9.611
	$k=5$	29	3	10.6718*	10.830	9.555
		28	2	6.1415	11.055	9.701
		27	1	4.9930	12.004	10.165
d. $k_0=3$	31	6	19.9904**	11.496	10.187	
	$k_1=3$	30	5	14.6115**	11.240	9.888
	$k=6$	29	4	9.8310*	10.995	9.658
		28	3	5.8154	10.896	9.618
		27	2	4.6816	11.050	9.679
		26	1	3.8273	11.991	10.122

**The corresponding observation is declared as an outlier at $\alpha = 0.05$.

*The corresponding observation is declared as an outlier at $\alpha = 0.10$.

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SHORT COMMUNICATIONS

ON THE DIOPHANTINE EQUATION $x^2 + 2^k = y^n$ II

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ABSTRACT. In [1], the authors made a conjecture that the diophantine equation $x^2 + 2^k = y^n$ where x, y, k, n are positive integers with $n \geq 3$ has exactly two families of solutions when k is even. In this paper this conjecture is proved.

1. INTRODUCTION

In 1992 Cohn [3] proved that the diophantine equation $x^2 + 2^k = y^n$ where $n \geq 3$ and k odd integer has exactly three families of solutions in positive integers x and y . When $k = 1$, the title equation was first solved by T. Nagell [6], and recently B. Sury [9] has given an elementary proof that this equation has no solution. The same problem with k even was considered later on by Arif and Abu Muriefah [1] and Cohn [4], and appeared to be of rather greater difficulty. In [1] authors made the following conjecture:

The diophantine equation

$$(1) \quad x^2 + 2^{2m} = y^n, \quad n \geq 3, m \geq 1$$

has exactly two families of solutions given by

$$x = 2^m, y^n = 2^{2m+1}, \quad \text{for all } m \geq 1$$

and

$$x = 11 \cdot 2^{3M}, \quad y = 5 \cdot 2^{2M}, \quad n = 3, \text{ for } m = 3M + 1.$$

In this paper we shall prove this conjecture. When $m = 1$, the conjecture is true by the result of [7]. It is also proved in [1] that equation (1) has no solution when n is even and $m > 1$. So we can suppose that n is an odd integer.

We start by the usual method of factorizing in the fields $\mathbb{Q}(i)$, and then we use a recent result of Bilu, Hanrot and Voutier [2] about primitive divisors of Lucas numbers. We start by giving some important definitions and the result proved in [2].

Definitions. A *Lucas pair* is a pair (α, β) of algebraic integers such that $\alpha + \beta$ and $\alpha\beta$ are non-zero co-prime rational integers and $\frac{\alpha}{\beta}$ is not a root of unity. Given a Lucas pair (α, β) one defines the corresponding *sequence* of Lucas numbers by

$$u_n(\alpha, \beta) = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \quad n \geq 0.$$

A prime number p is a primitive divisor of $u_n(\alpha, \beta)$ if p divides u_n but does not divide $(\alpha - \beta)^2 u_1 u_2 \dots u_{n-1}$.

Result (Theorem 1.4 of [2]). For $n > 30$, the n th term of any Lucas sequences has a primitive divisor.

For $5 \leq n \leq 30$ all values of pairs have been listed for which the n th term of Lucas sequences has no primitive divisors.

We first prove the following theorem.

Theorem 1. *If x, y and n are odd integers, then the diophantine equation (1) has a unique solution given by $x = 11, y = 5, n = 3$ and $m = 1$.*

Proof. It is enough to prove the result for $n = p$ an odd prime. Factoring equation (1) in $\mathbb{Q}(i)$, we get

$$(x + 2^m i)(x - 2^m i) = y^p$$

Since the factors on the left hand side are relatively prime and any unit of $\mathbb{Q}(i)$ is a p -th power, we can write

$$(2) \quad x + 2^m i = (a + bi)^p$$

where a and b are rational integers and $y = a^2 + b^2$. Since y is odd, a and b are of opposite parity. Let $\alpha = a + bi$ and $\bar{\alpha} = a - bi$. Then from (2) we get

$$\frac{\alpha^p - \bar{\alpha}^p}{\alpha - \bar{\alpha}} = \frac{2^m}{b}$$

Since the left hand side is an algebraic integer, this implies that $b|2^m$, hence b is a power of 2, hence a is odd. It is easy to check that $\bar{\alpha}/\alpha$ is not a root of unity because it is never equal to ± 1 or $\pm i$. Also $\gcd(\alpha + \bar{\alpha}, \alpha\bar{\alpha}) = \gcd(2a, a^2 + 2^{2m}) = 1$, so

$$u_t = \frac{\alpha' - \bar{\alpha}^t}{\alpha - \bar{\alpha}}, \quad t \geq 0$$

is a Lucas sequence. Since 2 divides $(\alpha - \bar{\alpha})^2$, therefore by definition 2 is not a primitive divisor. Using Theorem 1.4 of [2] we get $p \leq 30$, and by checking the tables of [2] we see that only $p = 3$ and $m = 1$ is possible. So equation (1) becomes $x^2 + 4 = y^3$ which has only one solution in x odd as $x = 11, y = 5$, [7]. This completes the proof of Theorem 1.

Proof of conjecture. From Theorem 1, it is sufficient to consider that x is even, so let

$$x = 2^\mu X, \quad y = 2^\gamma Y \quad \text{where } \mu > 0, \gamma > 0$$

and X, Y are odd integers. From (1) we obtain

$$(3) \quad 2^{2\mu} X^2 + 2^{2m} = 2^{n\gamma} Y^n$$

and therefore of three powers of 2, 2μ , $2m$, and $n\gamma$ which occur here, two must be equal and the third is greater. Thus we have three cases

1. $2\mu > 2m = n\gamma$, then from (3) we get

$$(2^{\mu-m} X)^2 + 1 = Y^n$$

and this equation is known to have no solutions [5].

2. $n\gamma > 2\mu = 2m$, then from (3), we get

$$(4) \quad X^2 + 1 = 2^{n\gamma-2\mu} Y^n$$

Considering modulo 8, we get $n\gamma - 2\mu = 1$ and so (4) becomes.

$$X^2 + 1 = 2Y^n$$

which has been proved to have only one $X = Y = 1$, [8], so that

$$x = 2^m \text{ and } y^n = 2^{2m+1}$$

3. $2m > 2\mu = n\gamma$, then

$$X^2 + 2^{2(m-\mu)} = Y^n$$

which by Theorem 1 has no solution in X odd as $X = 11, Y = 5, n = 3$ and $m - \mu = 1$ and accordingly $n\gamma = 2\mu$ implies $3|\mu$, so

$$X = 11.2^{3M}, \quad y = 5.2^{2M}, \quad n = 3 \quad \text{for} \quad m = 3M + 1.$$

This completes the proof of the conjecture.

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المجلة العربية للعلوم الرياضية

مجلة علمية محكمة تصدر نصف سنوياً عن الجمعية السعودية للعلوم الرياضية

بالتعاون مع قسم الرياضيات بجامعة الملك سعود

رصدان ١٤٢٢هـ—

الجلد السابع

ديسمبر / كانون أول ٢٠٠١م

العدد الثاني

هيئة التحرير

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قواعد النشر

١) تعني المجلة العربية للعلوم الرياضية بنشر البحوث الأصلية في جميع فروع العلوم الرياضية ، بما في ذلك الرياضيات البحتة والتطبيقية ، والإحصاء وبحوث العمليات ، والرياضيات ————— الحاسبة. يتكون كل مجلد من عددتين تصدر نصف سنوياً .

٢) تخضع جميع الأبحاث التي تقدم إلى المجلة لعملية تحكيم ، ويشترط فيها أنه لم يسبق نشرها أو ليست في طريقها إلى النشر لدى أية جهة أخرى. يقدم المؤلف (المؤلفون) البحث مطبوعاً بالشكل المتبع في أي عدد من أعداد المجلة ، ومبدئياً برقم التصنيف في أسفل الصفحة الأولى ، ويرسل للمجلة من ثلاث نسخ على العنوان التالي :

رئيس التحرير — المجلة العربية للعلوم الرياضية

قسم الرياضيات بجامعة الملك سعود

ص ٢٤٥٥ الرياض ١١٤٥١

المملكة العربية السعودية

٣) يعاد البحث بعد قبوله للنشر إلى المؤلف لإجراء التعديلات والتصحيحات المطلوبة ، ويفضل أن يصل البحث إلى المجلة في صيغته الأخيرة بواسطة البريد الالكتروني على هيئة TEX format .

٤) لا تتقاضى المجلة أي رسوم على نشر البحوث ، وتزود الباحث (الباحثين) بعشرين نسخة من كل بحث منشور .

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